

TEACH THE INTERFACE, KEEP THE DRAFTER: CROSS-MODEL SPECULATIVE DECODING

Anonymous authors

Paper under double-blind review

ABSTRACT

A feature-conditioned speculative drafter is usually coupled to the language model that supplied its training features. Can it be reused when the target changes size or family, without retraining the draft transformer? We study five linear maps that reconstruct the source model’s feature interface and fold into the drafter’s existing fusion projection. On Qwen3-8B, a Qwen3-4B drafter adapted with 16,384 examples and 25% token coverage retains 100.2% and 99.4% of native-drafter mean throughput on Math and GSM8K, but only 84.2% and 90.4% on Code and Chat. Fitting takes 24.9 minutes on two L40S GPUs, excluding data collection. Optimized concurrent serving reaches 1.025–1.160 times native aggregate throughput on the two mathematical workloads, with weaker transfer elsewhere. Reverse-size and cross-family transfers establish important limits rather than universal native parity. A controlled-input diagnostic and live weight-interpolation interventions support representation compatibility as a mechanism: moving token-trained maps halfway toward an MSE solution improves Code throughput by 37.6–39.9%. Yet AUF refinement improves GSM8K while degrading other workloads, and initialization and optimization affect the comparison. We provide complete recipe, cost, architecture and intervention results, separating native-relative throughput, acceptance and runtime metrics. The evidence supports inexpensive reuse of pretrained drafting computation under a measurable compatibility constraint.

1 INTRODUCTION

Speculative decoding replaces serial target-model generation with fast proposals followed by parallel verification (Leviathan et al., 2022; Chen et al., 2023). Feature-conditioned drafters such as DFlash (Chen et al., 2026) make this economical by consuming hidden states that the target has already computed. This creates a deployment dependency: a drafter trained on one model’s representations need not understand those of a different model, even when the vocabulary is shared. Rebuilding a drafter for every target spends adaptation compute and fragments a pretrained asset.

We investigate a narrow alternative: retain the entire source drafter, including its fusion, normalization, embeddings and vocabulary head, and learn how to translate the new target’s features into its expected interface. The source language model supplies paired supervision offline; it is absent from mapped inference. Five bias-free maps operate on corresponding layer slots. Their products with the original fusion are precomputed, so the deployed interface is one projection followed by the inherited normalization. The method does not introduce a new verifier or require an extra live source-model pass.

The central question is not simply whether a linear map fits hidden vectors. It is whether the resulting proposals preserve useful inference performance. We therefore connect three types of evidence: multi-workload timing and acceptance; adaptation cost versus data coverage; and interventions on representations that are subsequently evaluated in live decoding. Our primary transfer, T1, reuses the Qwen3-4B drafter for Qwen3-8B. T2 reverses this direction, T3 transfers within Llama, and T4 crosses both model family and tokenization. All four use the same selected fitting budget; architecture and loss exploration are concentrated on T1.

Three findings organize the paper. First, feature reconstruction is a strong practical recipe for T1, approaching native throughput on mathematical workloads without updating the draft body. Second, compatibility is workload-dependent: Code and Chat expose residual deficiencies, and the larger

transferred drafter in T2 does not beat the smaller native one. Third, the objective contrast is conditional. MSE outperforms the tested cold AUF and decaying-CE recipes, but they receive different supervision and initially use a lower learning rate. Warm AUF and an initialization intervention rule out an unconditional story that reconstruction is always the best endpoint.

This complements adapter specialization: when a target LoRA changes, an existing same-model interface can start from identity; cross-model transfer must first establish compatibility between distinct feature spaces. Our contribution is the measured boundary between these settings, a reproducible foldable interface, and evidence about why reconstruction is useful.

2 RELATED WORK

Speculative sampling preserves the intended target distribution through verification rather than trusting draft tokens (Leviathan et al., 2022; Chen et al., 2023). DFlash uses target features to condition parallel masked-block proposals (Chen et al., 2026). EAGLE-3 instead emphasizes token prediction and multi-layer feature fusion (Li et al., 2025). Our reconstruction objective translates the inputs of an already pretrained draft network; it is not a claim that feature prediction should replace token prediction when training a new drafter.

Efficient Draft Adaptation studies parameter- and data-efficient retargeting after target fine-tuning (Lin et al., 2026). LoRA provides a standard means of updating a small trainable factorization while retaining a pretrained backbone (Hu et al., 2021). The distinction here is that the frozen consumer expects features from another model, potentially of another width or vocabulary. Related RelaySpec experiments also examine linear interfaces (RelaySpec); we retain only their independently trained five-map confirmation as supplementary evidence, with its different batching and evaluation cohort explicitly separated.

AUF truncates masked-block CE after the first predicted failure (Yang & Li, 2026). We use that published objective as a comparator, alongside DFlash’s positional decay. Neither loss is a contribution of this work. The question is when a token objective can discover a useful cross-model interface through a frozen body, compared with direct reconstruction of the interface it already knows.

3 METHOD

3.1 FIVE MAPS BEFORE A FROZEN SOURCE FUSION

Let S denote the source language model whose pretrained drafter D_S is reused, and T the new target. At prefix position p , target layer slot i supplies $h_{pi}^t \in \mathbb{R}^{d_t}$; the source counterpart is $h_{pi}^s \in \mathbb{R}^{d_s}$. There are five slots, indexed $i = 1, \dots, 5$. Write the frozen source fusion as $F = [F_1, \dots, F_5]$, where $F_i \in \mathbb{R}^{d_s \times d_s}$, and let N_s denote its original RMSNorm (Zhang & Sennrich, 2019). We train

$$W_i \in \mathbb{R}^{d_s \times d_t}, \quad \hat{c}_p = N_s \left(\sum_{i=1}^5 F_i W_i h_{pi}^t \right), \quad c_p^s = N_s \left(\sum_{i=1}^5 F_i h_{pi}^s \right). \quad (1)$$

Both language models, the draft transformer, F , the RMSNorm gain, and the source embedding/head are frozen. The selected initialization is independently drawn Xavier-uniform matrices with seed 42. A rectangular identity is a separate ablation, not the selected method.

At export we form

$$G = [F_1 W_1, \dots, F_5 W_5] \in \mathbb{R}^{d_s \times 5d_t}. \quad (2)$$

The runtime applies $N_s(G[h_{p1}^t; \dots; h_{p5}^t])$. Five-map training adds no sequential mapper multiplication after folding. This does not imply equal cost to every native drafter: hidden width, draft-body capacity, kernel shapes and tokenizer conversion can differ. T1 trains 52,428,800 entries; interface-only adaptation is not synonymous with a tiny parameter count.

3.2 PAIRED FEATURE RECONSTRUCTION

The target generates each training response once. Both frozen models then process the same saved prefix causally; they do not independently generate different responses. With different tokenizers,

pairing instead uses eligible common text-prefix endpoints. Features are computed with their full causal context before persistence selects positions.

For $\epsilon_D = 10^{-6}$, define relative squared error

$$D(u, v) = \frac{\|u - v\|_2^2}{\|v\|_2^2 + \epsilon_D}. \quad (3)$$

For example b and its fixed selected positions S_b , the objective is

$$L_{\text{MSE}} = \frac{1}{B} \sum_{b=1}^B \frac{1}{|S_b|} \sum_{p \in S_b} \left[\frac{1}{5} \sum_{i=1}^5 D(W_i h_{bpi}^t, h_{bpi}^s) + D(\hat{c}_{bp}, c_{bp}^s) \right]. \quad (4)$$

Each example receives equal total weight. The denominator compares error with reference-vector magnitude, and the two terms constrain individual features and the normalized context consumed by the draft body. Reconstruction gradients stop at this interface; they do not traverse the draft transformer or vocabulary head.

3.3 TOKEN OBJECTIVES AND WARM REFINEMENT

For token training, choose 32 response anchors per example. Each block contains a clean anchor and 15 masked prediction slots. Target context is strictly before the anchor; attention is bidirectional within a block and absent between blocks. Offset j predicts the saved token $x_{b,a+j}$, for $j = 1, \dots, 15$. Define valid-label mask m_{baj} and CE $\ell_{baj} = -\log q_{baj}(x_{b,a+j})$. The microbatch loss is

$$L(w) = \frac{\sum_{b,a,j} m_{baj} w_{baj} \ell_{baj}}{\sum_{b,a,j} m_{baj} w_{baj}}, \quad w_{baj}^{\text{decay}} = e^{-(j-1)/7}, \quad w_{baj}^{\text{AUF}} = \text{stopgrad} \prod_{k < j} r_{bak}. \quad (5)$$

Here $r_{bak} = 1$ for a correct argmax or invalid label. AUF includes the first wrong token and removes subsequent labels; decay retains every valid label. Both backpropagate through the frozen draft body. We also refine an MSE checkpoint with AUF on overlapping or previously unseen examples, charging both training stages. Exact reduction and optimizer schedules appear in Appendix A.

4 EXPERIMENTAL DESIGN

Transfers and workloads. The arrow indicates the drafter’s source model $\rightarrow$ new target, not inference order:

Pair	Source $\rightarrow$ target	Each map shape
T1	Qwen3-4B $\rightarrow$ Qwen3-8B	2560×4096
T2	Qwen3-8B $\rightarrow$ Qwen3-4B	4096×2560
T3	Llama-3.1-8B $\rightarrow$ Llama-3.2-3B	4096×3072
T4	Qwen3-4B $\rightarrow$ Llama-3.1-8B	2560×4096

Each selected fit uses 16,384 NuminaMath-derived training examples and target-generated responses capped at 4,096 tokens. Evaluation uses four distinct 128-prompt cohorts: MATH test (Hendrycks et al., 2021), GSM8K test (Cobbe et al., 2021), LiveCodeBench (Jain et al., 2024), and UltraFeedback-derived instructions (Cui et al., 2023). These are separate workloads, not a pooled mixture. Canonical standalone evaluation is BF16 Transformers/DFlash on one L40S, greedy decoding, thinking disabled where supported, natural stopping and a 2,048-token response cap. This is a throughput evaluation; we do not report benchmark answer accuracy. Public-data provenance limitations and model pins are explicit in the appendix.

Fitting and selection. The main recipe uses 25% of prompt and response positions, three epochs, AdamW LR 10^{-3} , and the last checkpoint. Two GPUs with two examples per microbatch and two accumulation rounds give global batch eight. The T1 size/coverage study reuses the same saved cohort, with 4k/8k subsets nested in 16k. Half and quarter coverage are deterministic and fixed across epochs; quarter coverage is nested inside the half sample. Token objectives initially use

Table 1: Selected 16k/25% recipe, Transformers, 128 prompts per task, cap 2,048. All TPS values are arithmetic means of request rates. All standalone AR controls are complete; a dash denotes the unavailable native drafter for T3.

Pair	Task	AR TPS	Native TPS	Mapped TPS	Map / AR	Map / native	Native τ	Mapped τ
T1	math	37.84	219.79	220.26	5.821	1.0022 $\times$	8.2108	7.9841
T1	gsm	37.76	177.94	176.87	4.684	0.9940 $\times$	6.4788	6.2416
T1	code	37.50	151.63	127.61	3.403	0.8416 $\times$	5.6441	4.7391
T1	chat	35.09	89.50	80.93	2.306	0.9043 $\times$	3.3236	2.8868
T2	math	39.42	224.52	216.37	5.489	0.9637 $\times$	8.0911	7.6596
T2	gsm	39.72	177.32	169.53	4.268	0.9561 $\times$	6.3415	5.9874
T2	code	39.60	171.68	130.83	3.304	0.7621 $\times$	6.2619	4.8890
T2	chat	40.48	95.34	81.16	2.005	0.8512 $\times$	3.3608	2.8289
T3	math	61.90	—	145.43	2.350	—	—	4.7798
T3	gsm	64.45	—	140.60	2.182	—	—	4.1593
T3	code	63.09	—	118.57	1.879	—	—	3.4400
T3	chat	64.72	—	104.43	1.613	—	—	2.9735
T4	math	41.00	130.07	121.94	2.974	0.9374 $\times$	5.1211	4.6341
T4	gsm	41.11	127.35	90.84	2.210	0.7133 $\times$	4.4245	3.2257
T4	code	40.79	122.44	80.60	1.976	0.6583 $\times$	4.5230	3.0537
T4	chat	40.96	114.42	60.52	1.478	0.5289 $\times$	3.9538	2.0640

LR 10^{-4} and 32 response anchors. Thus we compare complete recipes, not losses under identical information and optimization. The evaluation cohorts have been inspected repeatedly during recipe selection; they are training holdouts, not untouched confirmatory tests. Most settings have one seed and one timing pass.

Metrics. For request b , let o_b be returned tokens and t_b measured latency. We distinguish

$$\bar{v} = \frac{1}{n} \sum_b \frac{o_b}{t_b}, \quad v_{\text{pool}} = \frac{\sum_b o_b}{\sum_b t_b}, \quad R = \frac{\bar{v}_{\text{mapped}}}{\bar{v}_{\text{native}}}, \quad \rho = 1 - \frac{\sum_b t_b^{\text{mapped}}}{\sum_b t_b^{\text{native}}}. \quad (6)$$

R is native-throughput retention, not retention of the improvement over AR. Acceptance τ pools emitted block lengths per verification step, including the target contribution and terminal truncation. It is not answer accuracy. Concurrent serving instead uses returned tokens divided by workload wall time, including drain. We never substitute one denominator for another.

AR timing controls. Saved T1 AR measurements are available: mean TPS is 37.84/37.76/37.50/35.09 across Math/GSM8K/Code/Chat; Table 6 reports these controls and the descriptive AR-relative ratios. Complete saved T3 AR rates are 61.90/64.45/63.09/64.72 TPS across the same four tasks; Table 1 includes their descriptive ratios.

5 RESULTS: USEFUL REUSE WITH A WORKLOAD BOUNDARY

T1 retains mathematical-workload throughput. The selected 16k/25% mapper reaches 220.26 tokens/s on Math and 176.87 on GSM8K, retaining 1.0022 and 0.9940 times native mean TPS (Table 1). Its acceptance is slightly lower than native: 7.984 versus 8.211 on Math. Comparable speed therefore need not mean identical proposal quality; the two pretrained drafters have different execution costs. The Code and Chat ratios fall to 0.8416 and 0.9043. Feature compatibility is sufficient for useful reuse here, but does not preserve every domain’s native performance.

The reverse and cross-family transfers constrain generality. T2 retains 0.9637/0.9561 on Math/GSM8K and 0.7621/0.8512 on Code/Chat. A larger source drafter is not automatically advantageous for a smaller target. T4 reaches 0.9374 on Math but only 0.5289 on Chat, including tokenizer-bridge costs. T3 demonstrates working proposals with mean rates 104.43–145.43 tokens/s, but no native-drafter comparator is established. These findings do not support a claim of universal acceleration across all model pairs.

Table 2: T1 size and coverage. Task cells: mean-TPS ratio to native / pooled acceptance. Three epochs, LR 10^{-3} , Xavier maps; final checkpoint. GM gives equal weight to tasks and is not mixed-traffic throughput.

Examples	Tokens	Fit min	Math	GSM8K	Code	Chat	GM
16,384	25%	24.91	1.0022 / 7.984	0.9940 / 6.242	0.8416 / 4.739	0.9043 / 2.887	0.9331
16,384	50%	37.21	0.9989 / 7.989	0.9848 / 6.254	0.8477 / 4.752	0.9053 / 2.888	0.9321
8,192	100%	53.90	0.9979 / 7.999	0.9901 / 6.224	0.8283 / 4.697	0.8914 / 2.819	0.9241
8,192	50%	17.79	0.9986 / 7.978	0.9753 / 6.257	0.8322 / 4.691	0.8882 / 2.812	0.9211
8,192	25%	11.59	0.9953 / 7.973	0.9686 / 6.204	0.8309 / 4.690	0.8749 / 2.802	0.9150
4,096	100%	18.30	0.9971 / 7.971	0.9797 / 6.187	0.7785 / 4.434	0.8537 / 2.681	0.8977
4,096	25%	6.06	0.9974 / 7.975	0.9756 / 6.169	0.7623 / 4.367	0.8406 / 2.635	0.8886
4,096	50%	9.56	0.9764 / 7.969	0.9422 / 6.165	0.7522 / 4.422	0.8096 / 2.662	0.8652

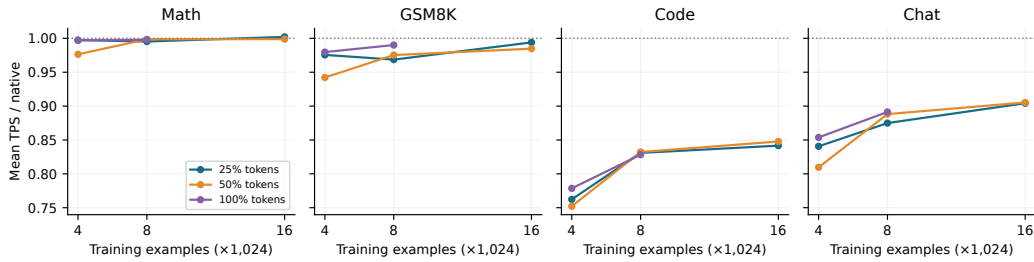


Figure 1: T1 coverage and example-count sensitivity. Points are measured runs; lines connect observations rather than fit a scaling law. Timing is from separate passes; no unmeasured confidence interval is implied.

More coverage is not automatically worth its fitting cost. At 16k, quarter coverage takes 24.91 minutes versus 37.21 for half coverage, a 33% reduction. Acceptance changes by less than 0.3% on every task. At 8k, complete-token reconstruction takes 53.90 minutes versus 17.79 for half coverage without a consistent acceptance improvement. Increasing examples helps Code/Chat more clearly than increasing coverage. The nonmonotone timing points, especially at 4k, caution against a fitted power-law interpretation. Our choice of 16k/25% is a practical point on this measured tradeoff, not a universal optimum.

Architecture and objective are both material. At 8k, MSE50 reaches 0.9986/0.9753/0.8322/0.8882 native-relative mean TPS across the four tasks. AUF reaches 0.8837/0.9107/0.3850/0.5578 and decay CE 0.8858/0.9187/0.4060/0.5733. Under the same token objectives, a joint map offers little Math improvement; direct-fusion training reaches only about 0.365 on Math and 0.222–0.223 on Code. Low-rank whole-map replacements underperform full five-map training. These controls change optimization geometry or the initial function; they are not evidence that an identity-preserving low-rank residual must fail. Updating the larger native drafter body gives the best selected architecture-control rates, including 1.0502 on Math with AUF. It is a different backbone and does not invalidate the narrower benefit of reusing the smaller frozen drafter. All architecture rows, fitting times and native EAGLE-3 controls are in Appendix D.

6 MECHANISM: RECONSTRUCTING A USABLE INTERFACE

Compatibility improves actual predictions, not only the training metric. On training-feature diagnostics, MSE has response context error 0.136 versus 0.940 for AUF and 0.900 for decay; response layer cosines are 0.944–0.985 versus 0.121–0.267 for AUF. This is expected from reconstruction training. More discriminating evidence uses the same held-out prefixes and block anchors across all maps. On Code, MSE improves full CE from 5.508 to 2.604 and correct prefix from 1.422 to 4.875 relative to AUF. On Chat, CE changes from 6.209 to 3.807 and prefix from 0.844 to 2.906. These improvements agree with the full live workload ranking.

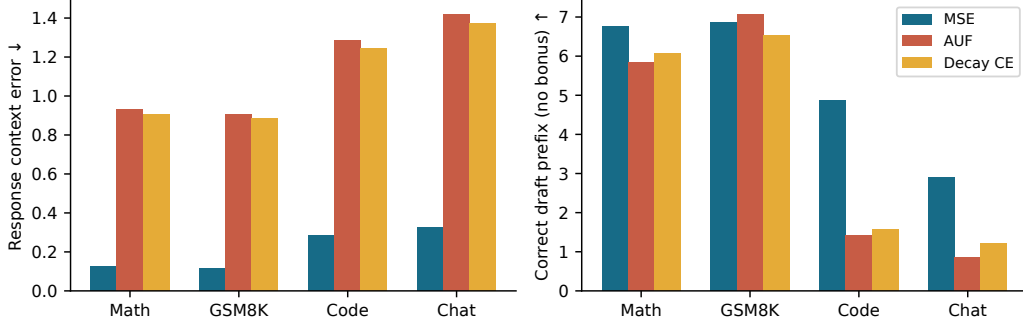


Figure 2: Common-input probe: eight held-out examples and 64 identical draft blocks per task per variant. Context error is reconstruction; correct prefix is offline token agreement without a verifier bonus. Neither is live acceptance or measured latency.

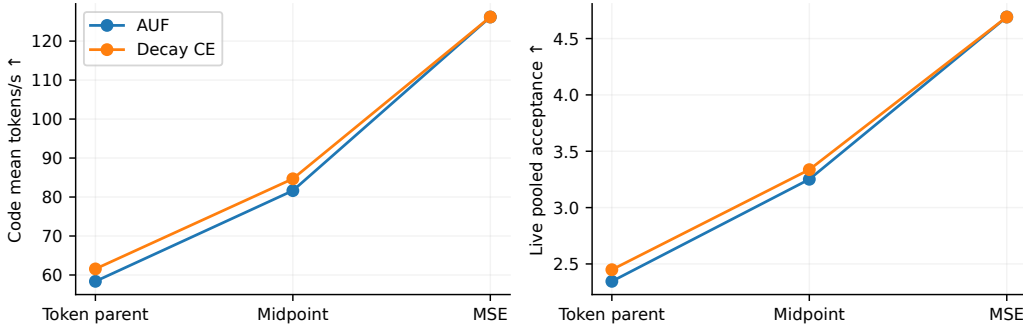


Figure 3: Live Code interventions, 128 prompts each. The midpoint averages all five token-trained and MSE maps without retraining. It improves throughput and acceptance for both token objectives, without reaching MSE. Endpoints were measured in separate passes.

Moving toward reconstruction improves live decoding. Set $W_i^{1/2} = (W_i^{\text{token}} + W_i^{\text{MSE}})/2$ and change nothing else. Code mean TPS rises from 58.37 to 81.64 for AUF and from 61.56 to 84.68 for decay: factors 1.3986 and 1.3756. Acceptance rises from 2.346 to 3.251 and 2.449 to 3.337. Summed latency falls 27.53% and 26.39%, respectively. Math also improves, by 6.48% and 7.25% in mean TPS. This intervention supports a useful direction toward the MSE solution, but changes all branches and their scales together; it does not isolate one geometric cause.

Warm token training exposes a tradeoff, not a universal ranking. From the 8k/MSE50 parent, AUF refinement on 4k/8k seen or unseen examples improves GSM8K but reduces Math, Code and Chat throughput. Unseen-8k refinement reaches 1.0606 times native GSM8K throughput with acceptance 6.622, at 47.42 minutes total fitting. The same fit gives 0.7051 on Code, below its MSE parent’s 0.8322. Warm seen-8k improves the small Math probe’s correct prefix from 6.766 to 7.563, yet its full live throughput remains below the MSE parent. An offline diagnostic is not a substitute for a workload measurement.

Initialization helps token recipes but does not explain away the gap. A separate retry starts all maps at $[I_{2560} \ 0]$, with 8,192 examples and LR 10^{-3} . Relative to the old AUF/decay recipes, both initialization and LR change. Code/Chat mean TPS improves roughly 19–25%, but both remain below MSE25 acceptance on every task. The MSE comparison changes only initialization: measured TPS declines 2.31–5.15%, while Math/GSM acceptance is almost unchanged and Code/Chat acceptance falls. In the retry, AUF is faster than MSE on GSM8K despite lower acceptance. Thus acceptance alone does not determine timing, and the experiment cannot assign the token-recipe gain separately to LR or initialization.

Table 3: Optimized vLLM, 128 prompts per workload. Panel A reports mapped throughput divided by native or AR throughput, as labelled. All AR cells are complete. Panel B preserves T3 absolute throughput because it has no native drafter.

A. Mapped throughput ratios

Pair	Clients	Math		GSM8K		Code		Chat	
		/ native	/ AR	/ native	/ AR	/ native	/ AR	/ native	/ AR
T1	1	1.058	6.346	1.053	4.959	0.917	3.853	0.929	2.320
T1	8	1.051	5.703	1.113	4.371	0.944	3.705	0.967	2.149
T1	16	1.076	4.771	1.025	3.742	0.886	3.065	0.977	1.919
T1	32	1.157	3.909	1.160	2.661	0.941	2.336	0.944	1.563
T2	16	0.828	3.868	0.974	3.014	0.705	2.427	0.758	1.519
T3	16	—	2.534	—	3.082	—	1.712	—	1.771
T4	16	0.552	1.876	0.630	2.163	0.491	1.088	0.364	1.008

B. T3 mapped throughput (tokens/s)

T3	16	2856.085	2241.589	2077.617	1899.952
----	----	----------	----------	----------	----------

The learned update is used, but is not a tiny universal correction. Live interventions on the AUF parent show that restoring the last map to its random initialization reduces Math mean TPS to 0.3643 times the parent; restoring the first costs much less. Removing 32 fusion-sensitive input directions reduces TPS to 0.1978 of the parent, whereas removing 32 random directions retains 0.9983. Keeping only rank two of each learned map update leaves acceptance 1.078, and even rank 128 retains only 0.7902 of parent TPS. These are not MSE compression results. Perpendicular-only correction retains 0.8029 of matched-control TPS; radial-only correction retains 0.1458. The reference direction is the random initial fused context, so this does not imply that radial changes are universally useless. Complete spectra and all live interventions appear in Appendix F.

7 DEPLOYMENT ACCOUNTING AND SCOPE

For an approximate stationary decoding regime, let c_D and c_V be per-step draft and verification time. Decode throughput is approximately $\tau/(c_D + c_V)$; end-to-end latency additionally includes prefill, stopping and host overhead. Thus lower acceptance can coexist with higher speed if draft computation is sufficiently cheaper. With a fraction f of original latency improved by factor s , Amdahl’s relation is $S = 1/[(1 - f) + f/s]$. We do not infer f from acceptance or claim measured multi-client gains from standalone results.

The separate optimized vLLM study (Kwon et al., 2023) uses the same selected checkpoints, four workloads, 128 requests per cell, and clients 1/8/16/32 for T1 and 16 for T2–T4. The profile uses BF16, FlashAttention, 15 proposals, compilation mode 3, automatic async scheduling, prefix caching reset before each cell and a 32,768-token batch budget. T4 uses the required cross-tokenizer bridge and a different model-runner generation; its overhead remains timed. All 80 method cells, including 28 AR baselines, are complete (Table 3). T1 Math/GSM8K retain 1.025–1.160 times native aggregate throughput across the tested client counts, while Code/Chat retain 0.886–0.977. At 16 clients, T2 retains 0.705–0.974 and T4 0.364–0.630. All T1–T4 AR serving measurements are complete; the tables report their throughput ratios.

8 LIMITATIONS AND CONCLUSION

The experiments isolate a practical reuse strategy more strongly than they isolate a loss function. MSE has paired-source supervision over prompt and response positions; token objectives use response blocks, another learning rate, and gradients through a frozen body. The identity retry is not a complete factorial control. Most results use one seed, and repeatedly inspected 128-prompt cohorts limit generalization and statistical confidence. Public dataset reconstruction is less fully pinned than the saved-token artifacts, and the Llama drafter was pretrained at a different block length. Native-

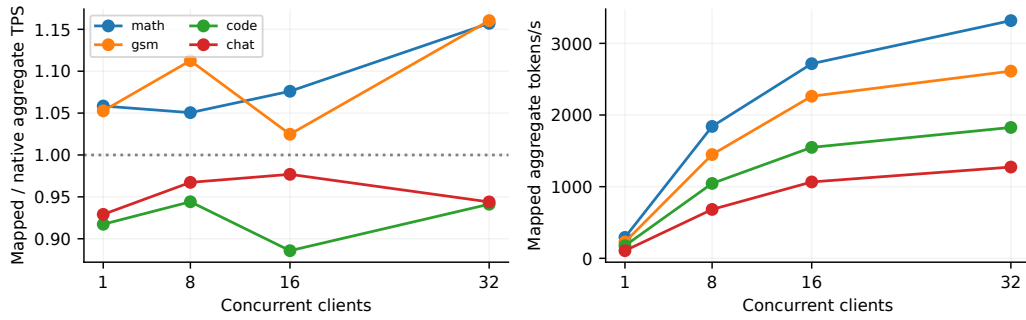


Figure 4: T1 optimized concurrent serving. Left: mapped/native aggregate throughput. Right: mapped aggregate throughput. Each point contains 128 requests, includes workload drain, and is a single timing pass.

body and EAGLE controls use other pretrained assets and cannot establish a matched-pretraining comparison.

Within those boundaries, a folded linear interface can reuse a pretrained drafter across target models at modest fitting cost. Mathematical workloads approach native throughput; other workloads reveal missing compatibility and tokenizer costs. Reconstruction, live interpolation, warm refinement and destructive interventions jointly support a concrete interpretation: a frozen drafter first needs a usable feature interface, after which task-directed optimization can help or harm according to workload. This is the basis for extending shared-drafter adaptation beyond same-model LoRA changes, without claiming that all target changes require the same objective.

REFERENCES

- Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating Large Language Model Decoding with Speculative Sampling. *arXiv preprint arXiv:2302.01318*, 2023. URL <https://arxiv.org/abs/2302.01318>.
- Jian Chen, Yesheng Liang, and Zhijian Liu. DFlash: Block Diffusion for Flash Speculative Decoding. *arXiv preprint arXiv:2602.06036*, 2026. URL <https://arxiv.org/abs/2602.06036>.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. URL <https://arxiv.org/abs/2110.14168>.
- Ganqu Cui, Lifan Yuan, Ning Ding, Guanming Yao, Wei Zhu, Yuan Ni, Guotong Xie, Zhiyuan Liu, and Maosong Sun. Ultrafeedback: Boosting language models with scaled ai feedback. *arXiv preprint arXiv:2310.01377*, 2023. URL <https://arxiv.org/abs/2310.01377>.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021. URL <https://arxiv.org/abs/2103.03874>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*, 2021. URL <https://arxiv.org/abs/2106.09685>.
- Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida I. Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. Livecodebench: Holistic and contamination free evaluation of large language models for code. *arXiv preprint arXiv:2403.07974*, 2024. URL <https://arxiv.org/abs/2403.07974>.

-
- 432 Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E.
433 Gonzalez, Hao Zhang, and Ion Stoica. Efficient Memory Management for Large Language Model
434 Serving with PagedAttention. *arXiv preprint arXiv:2309.06180*, 2023. URL <https://arxiv.org/abs/2309.06180>.
435
436 Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast Inference from Transformers via Specula-
437 tive Decoding. *arXiv preprint arXiv:2211.17192*, 2022. URL [https://arxiv.org/abs/](https://arxiv.org/abs/2211.17192)
438 [2211.17192](https://arxiv.org/abs/2211.17192).
439
440 Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE-3: Scaling up Inference Ac-
441 celeration of Large Language Models via Training-Time Test. *arXiv preprint arXiv:2503.01840*,
442 2025. URL <https://arxiv.org/abs/2503.01840>.
443
444 Luxi Lin, Zhihang Lin, Zhanpeng Zeng, Yuhao Chen, Qingyu Zhang, Jixiang Luo, Xuelong Li, and
445 Rongrong Ji. Efficiently Aligning Draft Models via Parameter- and Data-Efficient Adaptation.
446 *arXiv preprint arXiv:2603.09527*, 2026. URL <https://arxiv.org/abs/2603.09527>.
447
448 NuminaMath contributors. Numinamath-cot, 2024. URL <https://huggingface.co/datasets/AI-MO/NuminaMath-CoT>.
449
450 RelaySpec. Relayspec: Reproducible cross-target speculative decoding experiments. <https://github.com/AryamaVMurthy/RelaySpec>, 2026. Experiment artifact, revision
451 69f0e1f70e8c91437f61fb79aa84e9e4eb57118d.
452
453 Tianjian Yang and Meng Li. Spec-AUF: Accept-Until-Fail Training under Train-Inference Mis-
454 alignment for Masked Block Drafters. *arXiv preprint arXiv:2607.01893*, 2026. URL <https://arxiv.org/abs/2607.01893v1>.
455
456 Biao Zhang and Rico Sennrich. Root Mean Square Layer Normalization. *arXiv preprint*
457 *arXiv:1910.07467*, 2019. URL <https://arxiv.org/abs/1910.07467>.
458
459
460
461
462
463
464
465
466
467
468
469
470
471
472
473
474
475
476
477
478
479
480
481
482
483
484
485

Output agreement. In the selected Transformers comparisons with native-drafter controls (T1, T2 and T4), native and adapted drafters matched all 128 output sequences per task. In vLLM, token sequences diverged, consistent with documented numerical and batching variability. Semantic similarity does not imply identical token counts. See the [vLLM batch-invariance documentation](#).

A REPRODUCIBLE TRAINING AND EVALUATION CONTRACT

A.1 MODELS, LAYERS AND PARAMETER OWNERSHIP

All main transferred models train five maps, retain the source fusion and RMSNorm gain, and retain the source draft body, embedding and head. Source/target widths and direction are defined in the main text. Qwen3 models use source and target slots [1,9,17,25,33]. T3 uses source [1,8,15,22,29] and target [1,7,13,19,25]. T4 uses Qwen source slots and Llama target slots [1,8,15,22,29]. Indices are zero-based transformer block outputs; a library’s hidden-state tuple may include the embedding and must not silently shift these positions.

Asset	Pinned revision or content hash
Qwen/Qwen3-4B	1cfa9a7208912126459214e8b04321603b3df60c
Qwen/Qwen3-8B	b968826d9c46dd6066d109eabc6255188de91218
Qwen3-4B DFlash	b74e3a329c4d963783143b1e970d95b002be72bd
Qwen3-8B DFlash	9b41424b7109f9c5413454f481b09a82b85333f4
Unsloth Llama-3.1-8B-Instruct	4699cc75b550f9c6f3173fb80f4703b62d946aa5
Unsloth Llama-3.2-3B-Instruct	006f5dcd1393c3add266de40994ba96225e9689d

The inherited Llama drafter is content-pinned by SHA-256 738b12235328a8f3323b69e7d56bb7847c70908481fb86a209d12ca9d6cc1f3c. Its pretrained block size is 10; the selected evaluation uses block size 16, an explicit deviation from that pretraining contract. The Qwen mask ID is 151669; the Llama source draft uses 128002. T4 retains the Qwen draft mask and source vocabulary. There is no compatible pinned native drafter for the Llama-3.2-3B target in this study.

The native DFlash/PyTorch evaluation is based on DFlash 0.1.0, distribution hash b3fe38d9413efc30a952c8b38b34f1dcc696ba6d7b865acbf540a34cc53afaa2. The masked-block training implementation is the pinned SpecForge DFlash family implementation, with source files hashed in the accompanying manifest. The known training-source revision is 953d43a for the inherited DFlash training contract; local objective/integration changes are hashed separately. AUF is an implementation of the published support formula, not a claim of using author-released AUF training code. Full file hashes, per-pair model configurations and frozen source snapshots accompany the manuscript build.

A.2 DATA AND SAMPLING

The selected corpus consists of 16,384 saved target-generated NuminaMath-derived trajectories ([NuminaMath contributors, 2024](#)). T1 takes the same ordered saved cohort for all current size/coverage comparisons. A 4k or 8k fit takes its first 4,096 or 8,192 unique records. The normalized T1 cohort preserves token hashes, source row IDs and original shard hashes, but does not itself fully specify the public dataset revision. We therefore distinguish saved-token reproducibility from complete public-data regeneration; the unrelated earlier NuminaMath split is not silently substituted for this manifest.

T2/T3 reuse 4,096 compatible trajectories each and generate the remaining 12,288 with vLLM; the selected total remains 16,384. T2 contains 19,569,631 response tokens and T3 15,554,236. T3 retains the historical chat-template date (11 September 2026). The maximum-generation setting of the reused T3 subset was not recorded; new generation uses a 4,096-token cap. This is a provenance limitation, not evidence of independently generated 4,096-token responses for every example. Public dataset revision and full text-template provenance must be supplied with a release before claiming regeneration from the public source alone.

Let P_b and R_b be the prompt and response position sets. Half coverage samples $\lceil |P_b|/2 \rceil$ and $\lceil |R_b|/2 \rceil$ without replacement, independently by segment. Seeds derive from the token hash and

segment. Quarter coverage takes $\lceil |P_b|/4 \rceil$ and $\lceil |R_b|/4 \rceil$ positions from the corresponding half samples using a separate deterministic seed. Every token position is used at 100% coverage; padding is excluded. Sampling is fixed across epochs and both teachers use identical selected positions. Source and target process the full causal sequence before sampled features are persisted in BF16. Thus sampled supervision does not remove context from the teacher computation.

For T4, a source-token boundary and target-token boundary must represent the same eligible text prefix. Positions are selected at 25% of those eligible endpoints within each segment, not at 25% of arbitrary target tokens. At inference the bridge converts source proposals into target-token candidates and verifies with the target. Text conversion, source-prefix bookkeeping and synchronization are included in timing. T4 is not a vocabulary-identical test of only the linear maps.

Evaluation uses fixed 128-prompt MATH, GSM8K, LiveCodeBench and UltraFeedback instruction cohorts with target-appropriate saved prompt IDs. Chat is single-turn generation, not a multi-turn benchmark. No accuracy verifier is used to select outputs or score task success in these throughput tables. The same saved benchmark prompts recur across fitting comparisons; repeated inspection makes these development measurements even though the diagnostic examples are outside the training manifest. The accompanying hashes pin the available concrete workload files and their ordering.

Standalone timing boundary. After model loading, two separate warmup prompts generate up to 64 tokens each. Input tokenization and host-to-device placement precede timing. A GPU synchronization brackets the complete DFlash generation call; latency includes target prefill, draft construction, verification and decoding-loop overhead. Output transfer to CPU, text decoding and result serialization occur afterward. Requests are processed one at a time per GPU, including when the cohort is partitioned among workers. For Qwen, sampling arguments are temperature 0, top- p 1 and top- k 0; target and draft use PyTorch SDPA. The maximum context is 8,192 tokens, with an evaluation response cap of 2,048 and natural EOS stopping. No answer extraction or task-verifier pass enters the timed region.

A.3 OPTIMIZER AND EXACT REDUCTIONS

All selected fits initialize independently at seed 42, use three epochs, no development-loss checkpoint selection, and export the final checkpoint. Trainable maps, relative-MSE arithmetic and optimizer state use FP32. Draft computation for token objectives uses BF16 autocast and a frozen vocabulary head. The selected MSE matrix multiplications disable TF32. AdamW uses zero weight decay and global gradient-norm clipping at 1. The peak learning rate is 10^{-3} for MSE and 10^{-4} for the original AUF/decay recipes; the identity retry uses 10^{-3} for all three.

With K total updates, the learning rate for update $k \in \{1, \dots, K\}$ is

$$\eta_k = \eta_{\max} \min \left(1, \frac{k}{0.05K} \right) \frac{1 + \cos \left(\pi \max \left(0, \frac{k - 0.05K}{0.95K} \right) \right)}{2}. \quad (7)$$

For 4,096/8,192/16,384 examples and three epochs, $K = 1536/3072/6144$. Each epoch uses a deterministic permutation seeded by $42 + e$, with $e = 0, 1, 2$. Two GPUs each load two examples, repeated over two accumulation rounds. After each local loss is divided by two, DDP averages gradients over the two GPUs. Consequently

$$L_{\text{step}} = \frac{1}{4} \sum_{g=1}^2 \sum_{u=1}^2 L_{g,u}. \quad (8)$$

AdamW uses its default momentum coefficients (0.9, 0.999) and optimizer epsilon 10^{-8} . For MSE this is equal-example weighting over the eight records. For token objectives it is the mean of four locally token-normalized microbatch losses; it is not a single global token-normalized loss when support counts differ. CE is computed over the full vocabulary in two-block chunks, accumulating numerator and denominator before microbatch division. The denominator is clamped below by the floating-point type’s smallest positive normal value; all supervised blocks here have positive support. Padding, the clean anchor and nonsupervised labels have zero loss.

Complete selected training loop.

1. Freeze both teachers and the source drafter. Generate and persist each target response once; preserve prompt/response lengths and token IDs.
2. Causally extract the five source/target feature sequences, align token or text-prefix positions, and persist the fixed selected BF16 feature rows.
3. Initialize the five FP32 maps. For each deterministic eight-record update, compute Eq. 4 with equal-example weighting, accumulate four local losses, average gradients, clip, set η_k , and update only the maps.
4. For AUF/decay comparators, sample response anchors, build the clean-anchor plus mask blocks, compute Eq. 5, and backpropagate through the frozen draft transformer to the trainable interface. Recompute detached AUF support at each forward pass.
5. After three epochs, form Eq. 2, cast the folded export to BF16, preserve source embeddings/head and normalization, and evaluate the final checkpoint. Warm refinement instead loads the MSE epoch-three maps, starts a fresh AUF optimizer/schedule, and trains three further epochs.

A.4 FITTING VERSUS COLLECTION COST

Transfer	Fit minutes	Fit GPU-hours	Updates	Last minibatch MSE
T1	24.91	0.8303	6144	0.239890
T2	19.09	0.6363	6144	0.270358
T3	20.58	0.6861	6144	0.421100
T4	20m33s fit allocation; optimizer-only time not substituted			

T4’s allocation duration includes setup and is not comparable with the other optimizer-loop times. Last-minibatch MSE is not an epoch mean or held-out loss. A shared earlier T1 16k feature extraction took 3,325.78 seconds on one GPU; earlier 4k T2/T3 captures took 1,032.78/890.90 seconds. These are specific recorded captures, not the cost of the entire final adaptation campaign. One cache serves multiple fits and should be charged once. None of the 24.9-minute fitting claim includes rollout generation, extraction, setup, export or evaluation.

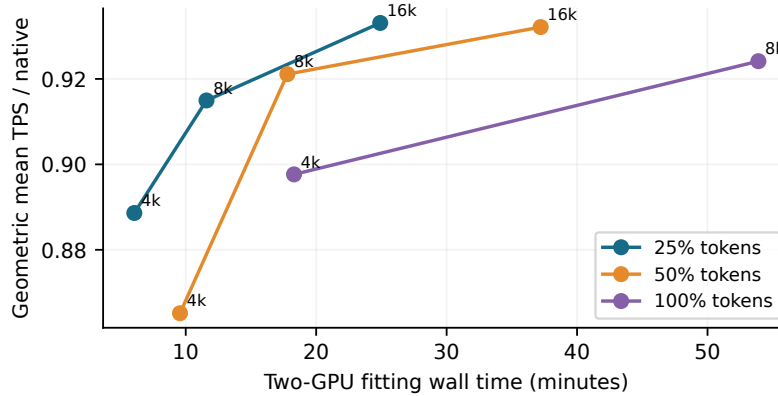


Figure 5: Measured T1 adaptation tradeoff, excluding collection. Labels give example counts in units of 1,024. The vertical quantity is the equally weighted geometric mean of four standalone TPS ratios, not a measured mixture throughput.

Table 4: Recorded shared feature extraction costs. These stages are not multiplied by the number of fits.

Cache	Examples	Seconds	GPU hours
T2	4096	1032.78	0.2869
T1	16384	3325.78	0.9238
nanocoder	4096	688.36	0.1912

Continued

Cache	Examples	Seconds	GPU hours
math	4096	642.81	0.1786
T3	4096	890.90	0.2475
kicad	4096	4048.77	1.1247

B ANALYTICAL INTERPRETATION WITH EXPLICIT ASSUMPTIONS

B.1 WHY INTERFACE ERROR CAN MATTER FOR TOKEN DECISIONS

Stack all context features that a particular draft block consumes into C , and write its frozen downstream logit function as $f(C)$. Suppose f is locally L -Lipschitz in a stated neighborhood from Euclidean context norm to maximum logit error. Then

$$\|f(\hat{C}) - f(C_s)\|_\infty \leq L \|\hat{C} - C_s\|_2. \quad (9)$$

If the source-context logit winner has margin m over the next token and $m > 2L \|\hat{C} - C_s\|_2$, its argmax is unchanged. The proof applies the error bound once to the winning logit and once to the competing logit. This is a sufficient local condition, not a measured bound in these experiments. Preserving the source drafter’s decision also does not guarantee agreement with the new target. Mean reconstruction loss does not control every position or supply a universal lower bound on m . The result explains why compatibility is a plausible mechanism, not why MSE must maximize acceptance.

For a block with K proposals, let $L_a \in \{0, \dots, K\}$ be the number of consecutive accepted draft tokens. The identity

$$\mathbb{E}[L_a] = \sum_{j=1}^K \Pr(L_a \geq j) \quad (10)$$

follows by writing L_a as a sum of threshold indicators; it requires no independence assumption. AUF focuses supervision on this prefix support. However, when a random interface predicts almost no correct leading tokens, the supported fraction can be near $1/K$. In the held-out initializer diagnostic it is 6.81–7.48% for $K = 15$. This is a post-hoc diagnostic of support, not a measured training-gradient history, and does not prove that AUF is unable to learn from that initialization.

B.2 THE FACTORIZATION CHANGES OPTIMIZATION EVEN WHEN EXPORTS HAVE THE SAME SHAPE

For a branch with effective matrix $G_i = F_i W_i$ and fixed F_i , the chain rule gives

$$\nabla_{W_i} L = F_i^\top \nabla_{G_i} L. \quad (11)$$

For one plain SGD step of size η , this induces

$$\Delta G_i = -\eta F_i F_i^\top \nabla_{G_i} L. \quad (12)$$

A directly optimized G_i instead takes $-\eta \nabla_{G_i} L$. Thus the pretrained fusion changes the optimization geometry of five-map training. If F_i is nonsingular, the effective linear function class need not be smaller; if singular, accessible outputs are constrained to its range. The actual experiments use AdamW and different objectives, so the SGD relation is explanatory algebra, not an exact account of their optimizer trajectories or proof that conditioning caused the observed gap.

B.3 ENERGY IS NEITHER VARIANCE NOR ACCEPTANCE RETENTION

For an update matrix A with singular values σ_j , the recorded energy is $E_k = \sum_{j \leq k} \sigma_j^2 / \sum_j \sigma_j^2$. Activation correction energy is measured on uncentered vectors, so a mean offset contributes to it. With feature second moment $S = \mathbb{E}[hh^\top]$, residual update $A - A_k$ has mean squared action

$$\mathbb{E} \|(A - A_k)h\|_2^2 = \text{tr}((A - A_k)S(A - A_k)^\top). \quad (13)$$

Matrix energy ignores S , and feature energy ignores downstream decision margins. Neither is a throughput guarantee. Our AUF rank truncations directly test deployment and show severe losses even though some activation spectra are concentrated.

For RMSNorm, write $N(z) = \Gamma z/r$, where $r = \sqrt{\|z\|^2/d + \epsilon_N}$ and Γ is diagonal gain. Its Jacobian is

$$J_N(z) = \Gamma \left(\frac{I}{r} - \frac{zz^\top}{dr^3} \right). \quad (14)$$

When $\epsilon_N = 0$, the radial direction is locally annihilated. Finite corrections relative to a random z_0 need not behave like small positive rescalings; radial removal can change the normalized direction of a combined update. This is why the radial/perpendicular interventions cannot establish that the parallel component of a trained map is globally useless.

B.4 SPEED, RETENTION AND AMORTIZATION

Native throughput retention $R = v_m/v_n$ is not the fraction of AR-relative savings retained. For a common-work denominator, a latency-savings retention would be $(T_{AR} - T_m)/(T_{AR} - T_n)$, which requires a valid AR reference. These are not interchanged in our tables. If adaptation costs C GPU-seconds and saves δ GPU-seconds per deployed request relative to a declared feasible alternative, break-even requires $N\delta > C$. When mapped decoding is slower than an already available native drafter, $\delta < 0$ for that comparator and there is no inference-time payback against it. Reuse can still avoid the cost of training or maintaining another drafter, but that avoided pretraining cost is not measured here.

C ALL SIZE, COVERAGE AND OBJECTIVE MEASUREMENTS

All following standalone rows use complete 128-prompt cohorts. Results from incomplete cohorts are not extrapolated. Quarter/half/full coverage refers to prompt and response feature supervision; the token-objective block-anchor budget is a different axis. No new main-study rollout-cap sweep was performed: all selected main fits use the recorded 4,096 cap. Earlier tiny-position reconstruction experiments are excluded from the approved coverage study.

Table 5: All completed standalone AR controls, 128 prompts per task. Mean TPS averages request rates; pooled TPS divides total output tokens by summed request time.

Pair	Task	Measured prompts	Status	Mean TPS	Pooled TPS
T1	math	128	complete	37.8373	37.7338
T1	gsm	128	complete	37.7622	37.7580
T1	code	128	complete	37.4970	37.4598
T1	chat	128	complete	35.0880	35.1556
T2	math	128	complete	39.4153	39.4552
T2	gsm	128	complete	39.7241	39.7216
T2	code	128	complete	39.6034	39.6920
T2	chat	128	complete	40.4786	40.5417
T3	math	128	complete	61.8956	62.0106
T3	gsm	128	complete	64.4510	64.4528
T3	code	128	complete	63.0876	63.1301
T3	chat	128	complete	64.7250	64.8830
T4	math	128	complete	41.0048	40.8312
T4	gsm	128	complete	41.1094	41.0486
T4	code	128	complete	40.7880	40.7296
T4	chat	128	complete	40.9593	40.9169

Table 7: Selected standalone pooled throughput. These denominators differ from the main mean-request TPS table. T4 native/mapped pooled values are not supplied by the final summary; none are inferred.

Pair	Task	AR pooled TPS	Native pooled TPS	Mapped pooled TPS	Map / AR
T1	math	37.7338	216.03	217.27	5.7580
T1	gsm	37.7580	173.86	173.75	4.6017
T1	code	37.4598	144.52	125.46	3.3492

Continued

Pair	Task	AR pooled TPS	Native pooled TPS	Mapped pooled TPS	Map / AR
T1	chat	35.1556	87.64	78.42	2.2307
T2	math	39.4552	222.28	212.85	5.3947
T2	gsm	39.7216	173.61	165.37	4.1632
T2	code	39.6920	174.01	136.92	3.4496
T2	chat	40.5417	93.64	79.74	1.9669
T3	math	62.0106	N/A	176.56	2.8473
T3	gsm	64.4528	N/A	149.88	2.3254
T3	code	63.1301	N/A	125.29	1.9846
T3	chat	64.8830	N/A	108.34	1.6698
T4	math	40.8312			
T4	gsm	41.0486			
T4	code	40.7296			
T4	chat	40.9169			

Table 8: Complete size/coverage measurements with reused T1 AR controls. Fitting cost is shared across tasks.

Examples	Coverage	Task	AR mean TPS	Mean TPS	Map/AR	Pooled TPS	Ratio/native	τ	Fit min
16384	50%	chat	35.0880	81.0312	2.3094	78.5978	0.9053	2.8882	37.21
16384	25%	chat	35.0880	80.9335	2.3066	78.4211	0.9043	2.8868	24.91
8192	100%	chat	35.0880	79.7790	2.2737	76.6755	0.8914	2.8186	53.90
8192	50%	chat	35.0880	79.4945	2.2656	76.5353	0.8882	2.8125	17.79
8192	25%	chat	35.0880	78.3080	2.2318	75.5334	0.8749	2.8017	11.59
4096	100%	chat	35.0880	76.4124	2.1777	73.0912	0.8537	2.6811	18.30
4096	25%	chat	35.0880	75.2389	2.1443	71.8594	0.8406	2.6346	6.06
4096	50%	chat	35.0880	72.4631	2.0652	69.8141	0.8096	2.6619	9.56
16384	50%	code	37.4970	128.5359	3.4279	125.8924	0.8477	4.7515	37.21
16384	25%	code	37.4970	127.6082	3.4032	125.4637	0.8416	4.7391	24.91
8192	50%	code	37.4970	126.1823	3.3651	124.0080	0.8322	4.6912	17.79
8192	25%	code	37.4970	125.9833	3.3598	124.0737	0.8309	4.6900	11.59
8192	100%	code	37.4970	125.5835	3.3492	123.7294	0.8283	4.6968	53.90
4096	100%	code	37.4970	118.0456	3.1481	117.1717	0.7785	4.4341	18.30
4096	25%	code	37.4970	115.5767	3.0823	115.8810	0.7623	4.3674	6.06
4096	50%	code	37.4970	114.0524	3.0416	114.0188	0.7522	4.4222	9.56
16384	25%	gsm	37.7622	176.8707	4.6838	173.7467	0.9940	6.2416	24.91
8192	100%	gsm	37.7622	176.1834	4.6656	173.1834	0.9901	6.2244	53.90
16384	50%	gsm	37.7622	175.2307	4.6404	172.5605	0.9848	6.2536	37.21
4096	100%	gsm	37.7622	174.3330	4.6166	171.3420	0.9797	6.1869	18.30
4096	25%	gsm	37.7622	173.5966	4.5971	170.9397	0.9756	6.1689	6.06
8192	50%	gsm	37.7622	173.5459	4.5958	170.7942	0.9753	6.2568	17.79
8192	25%	gsm	37.7622	172.3516	4.5641	169.7691	0.9686	6.2040	11.59
4096	50%	gsm	37.7622	167.6484	4.4396	164.8822	0.9422	6.1646	9.56
16384	25%	math	37.8373	220.2604	5.8212	217.2721	1.0022	7.9841	24.91
16384	50%	math	37.8373	219.5402	5.8022	216.8576	0.9989	7.9892	37.21
8192	50%	math	37.8373	219.4753	5.8005	216.7230	0.9986	7.9776	17.79
8192	100%	math	37.8373	219.3158	5.7963	216.8693	0.9979	7.9994	53.90
4096	25%	math	37.8373	219.2268	5.7939	216.1044	0.9974	7.9747	6.06
4096	100%	math	37.8373	219.1429	5.7917	215.9099	0.9971	7.9711	18.30
8192	25%	math	37.8373	218.7521	5.7814	216.4006	0.9953	7.9725	11.59
4096	50%	math	37.8373	214.6061	5.6718	212.3390	0.9764	7.9689	9.56

Table 9: T1 recipe comparison: 4k and 8k

Recipe	Examples	Fit min	Math	GSM8K	Code	Chat	Geomean $\times$ native
MSE 100%	8192	53.90	0.9979 $\times$ / 7.999	0.9901 $\times$ / 6.224	0.8283 $\times$ / 4.697	0.8914 $\times$ / 2.819	0.9241
MSE 50%	8192	17.79	0.9986 $\times$ / 7.978	0.9753 $\times$ / 6.257	0.8322 $\times$ / 4.691	0.8882 $\times$ / 2.812	0.9211
MSE 100%	4096	18.30	0.9971 $\times$ / 7.971	0.9797 $\times$ / 6.187	0.7785 $\times$ / 4.434	0.8537 $\times$ / 2.681	0.8977

Continued

Recipe	Examples	Fit min	Math	GSM8K	Code	Chat	Geomean $\times$ native
MSE 50%	4096	9.56	0.9764 $\times$ / 7.969	0.9422 $\times$ / 6.165	0.7522 $\times$ / 4.422	0.8096 $\times$ / 2.662	0.8652
DFlash decay CE	8192	28.24	0.8858 $\times$ / 6.947	0.9187 $\times$ / 5.693	0.4060 $\times$ / 2.449	0.5733 $\times$ / 1.812	0.6597
AUF	8192	31.64	0.8837 $\times$ / 6.892	0.9107 $\times$ / 5.677	0.3850 $\times$ / 2.346	0.5578 $\times$ / 1.767	0.6447
DFlash decay CE	4096	13.02	0.7912 $\times$ / 6.151	0.7585 $\times$ / 4.750	0.3324 $\times$ / 2.000	0.4823 $\times$ / 1.576	0.5569
AUF	4096	13.29	0.7741 $\times$ / 6.017	0.7415 $\times$ / 4.580	0.3180 $\times$ / 1.910	0.4687 $\times$ / 1.532	0.5408

The 4k complete-coverage fit was timed on a different L40S node from the corresponding half-coverage control. Separate timing passes, node placement and one-seed variation limit attribution of small differences to token coverage.

D ARCHITECTURE CONTROLS AND THEIR COSTS

Five-W is Eq. 1. Direct fusion trains $G \in \mathbb{R}^{2560 \times 20480}$ initialized to the same folded initial function. A joint map trains $J \in \mathbb{R}^{12800 \times 20480}$ before frozen F , initialized block-diagonally from the five initial maps. BA28/56/128 replace each entire W_i by $B_i A_i$, initialized by the rank- r SVD approximation of the Xavier map with square-root singular values split between factors. They are not low-rank residual updates around a retained full matrix. The native-body rank-128 LoRA uses alpha 256, dropout zero, and attention/MLP projections in the larger Qwen3-8B-native drafter; its fusion is excluded.

Five-W/direct fusion have 52,428,800 trainable entries; joint mapping has 262,144,000; BA28/56/128 have 931,840/1,863,680/4,259,840; the native-body control has 48,496,640. The architecture sweep evaluates Math and Code, while the reused five-W and native-body endpoints cover all four tasks. Absent GSM/Chat architecture-control cells were not run and are not inferred.

The EAGLE-3 control is a native external checkpoint under its pinned Transformers tree implementation; it is not a transferred EAGLE mapper trained in this campaign. Its emitted/verification accounting is EOS/cap-trimmed and overshoot work remains timed.

Table 10: Selected T1 architecture results

Cell	Task	TPS	$\times$ native	AR TPS ratio	Latency speedup $\times$ native	Emitted / verification	Fit min
T1_selected_body128_auf_n8192_e3	math	230.8243	1.0502	6.1004	1.0504	8.6097	31.79
T1_selected_body128_decay7_n8192_e3	math	226.1528	1.0290	5.9770	1.0273	8.4241	31.78
Native DFlash (reference)	math	219.7874	1.0000	5.8087	1.0000	8.2108	no fit
Five-W — MSE50	math	219.4753	0.9986	5.8005	1.0032	7.9776	17.79
Five-W — MSE100	math	219.3158	0.9979	5.7963	1.0039	7.9994	53.90
T1_selected_joint_decay7_n8192_e3	math	195.2239	0.8882	5.1596	0.8746	6.9644	33.29
Five-W — decay CE	math	194.6915	0.8858	5.1455	0.8689	6.9474	28.24
Five-W — AUF	math	194.2206	0.8837	5.1330	0.8645	6.8923	31.64
T1_selected_joint_auf_n8192_e3	math	193.3218	0.8796	5.1093	0.8673	6.9562	36.40
T1_selected_ba128_decay7_n8192_e3	math	149.9202	0.6821	3.9622	0.6564	5.2374	24.06
T1_selected_ba128_auf_n8192_e3	math	143.0149	0.6507	3.7797	0.6274	4.9728	29.49
T1_selected_ba56_decay7_n8192_e3	math	119.0842	0.5418	3.1473	0.5211	4.2197	24.61
T1_selected_ba56_auf_n8192_e3	math	109.9603	0.5003	2.9061	0.4765	3.7969	25.22
T1_selected_ba28_decay7_n8192_e3	math	93.2439	0.4242	2.4643	0.4070	3.2391	28.12

Continued

Cell	Task	TPS	× native	AR TPS ratio	Latency speedup × native	Emitted / verification	Fit min
T1_selected_ba28_auf_n8192_e3	math	82.5736	0.3757	2.1823	0.3621	2.8760	29.76
T1_selected_fusion_decay7_n8192_e3	math	80.2527	0.3651	2.1210	0.3530	2.8394	27.68
T1_selected_fusion_auf_n8192_e3	math	80.1810	0.3648	2.1191	0.3537	2.8402	26.90
T1_native_EAGLE3	math	72.1780	0.3284	1.9076	0.3421	3.6199	external check-point 31.79
T1_selected_body128_auf_n8192_e3	gsm	194.8627	1.0951	5.1603	1.0958	7.0986	31.78
T1_selected_body128_decay7_n8192_e3	gsm	187.6579	1.0546	4.9695	1.0519	6.8111	31.78
Native DFlash (reference)	gsm	177.9411	1.0000	4.7121	1.0000	6.4788	no fit
Five-W — MSE100	gsm	176.1834	0.9901	4.6656	0.9961	6.2244	53.90
Five-W — MSE50	gsm	173.5459	0.9753	4.5958	0.9823	6.2568	17.79
Five-W — decay CE	gsm	163.4700	0.9187	4.3289	0.9116	5.6926	28.24
Five-W — AUF	gsm	162.0495	0.9107	4.2913	0.9053	5.6773	31.64
T1_native_EAGLE3	gsm	75.5792	0.4247	2.0014	0.4492	3.6926	external check-point 31.79
T1_selected_body128_auf_n8192_e3	code	155.3287	1.0244	4.1424	1.0369	5.8636	31.78
T1_selected_body128_decay7_n8192_e3	code	153.8642	1.0148	4.1034	1.0164	5.7508	31.78
Native DFlash (reference)	code	151.6250	1.0000	4.0437	1.0000	5.6441	no fit
Five-W — MSE50	code	126.1823	0.8322	3.3651	0.8581	4.6912	17.79
Five-W — MSE100	code	125.5835	0.8283	3.3492	0.8562	4.6968	53.90
T1_selected_joint_decay7_n8192_e3	code	70.9838	0.4682	1.8931	0.5178	2.8271	33.29
T1_selected_joint_auf_n8192_e3	code	68.2183	0.4499	1.8193	0.5004	2.7418	36.40
T1_native_EAGLE3	code	63.8810	0.4213	1.7036	0.4656	3.3384	external check-point 28.24
Five-W — decay CE	code	61.5596	0.4060	1.6417	0.4484	2.4489	31.64
Five-W — AUF	code	58.3696	0.3850	1.5566	0.4283	2.3456	24.06
T1_selected_ba128_decay7_n8192_e3	code	43.8303	0.2891	1.1689	0.3157	1.7185	29.49
T1_selected_ba128_auf_n8192_e3	code	41.5391	0.2740	1.1078	0.2974	1.6302	24.61
T1_selected_ba56_decay7_n8192_e3	code	38.7609	0.2556	1.0337	0.2747	1.4981	25.22
T1_selected_ba56_auf_n8192_e3	code	37.3163	0.2461	0.9952	0.2623	1.4330	28.12
T1_selected_ba28_decay7_n8192_e3	code	35.8736	0.2366	0.9567	0.2503	1.3653	29.76
T1_selected_ba28_auf_n8192_e3	code	34.7612	0.2293	0.9270	0.2431	1.3335	26.90
T1_selected_fusion_auf_n8192_e3	code	33.8244	0.2231	0.9021	0.2360	1.2953	27.68
T1_selected_fusion_decay7_n8192_e3	code	33.6595	0.2220	0.8977	0.2353	1.2921	31.79
T1_selected_body128_auf_n8192_e3	chat	91.9628	1.0275	2.6209	1.0202	3.4139	31.78
T1_selected_body128_decay7_n8192_e3	chat	90.2557	1.0084	2.5723	1.0076	3.3477	no fit
Native DFlash (reference)	chat	89.5033	1.0000	2.5508	1.0000	3.3236	53.90
Five-W — MSE100	chat	79.7790	0.8914	2.2737	0.8749	2.8186	17.79
Five-W — MSE50	chat	79.4945	0.8882	2.2656	0.8733	2.8125	external check-point 28.24
T1_native_EAGLE3	chat	61.8191	0.6907	1.7618	0.7173	3.1783	31.64
Five-W — decay CE	chat	51.3131	0.5733	1.4624	0.5603	1.8115	
Five-W — AUF	chat	49.9253	0.5578	1.4229	0.5459	1.7667	

Table 6: Saved T1 Transformers AR controls and selected 16k/25% mapper, 128 prompts per task, cap 2,048. Rates are arithmetic means of request TPS. These are not vLLM concurrent-serving measurements.

Task	AR TPS	Native TPS	Mapped TPS	Native / AR	Mapped / AR
math	37.8373	219.7874	220.2604	5.8087	5.8212
gsm	37.7622	177.9411	176.8707	4.7121	4.6838
code	37.4970	151.6250	127.6082	4.0437	3.4032
chat	35.0880	89.5033	80.9335	2.5508	2.3066

Table 11: Selected T1 architecture results

Cell	Trainable parameters	Fit min	Fit GPU-hours	Setup s	Interface init s	Export s	Peak GiB
T1_selected_fusion_auf_n8192_e3	52,428,800	26.90	0.90	11.94	0.10	1.32	5.47
T1_selected_ba28_auf_n8192_e3	931,840	29.76	0.99	7.24	1.72	1.01	4.52
T1_selected_ba56_auf_n8192_e3	1,863,680	25.22	0.84	8.19	1.79	1.10	4.54
T1_selected_ba128_auf_n8192_e3	4,259,840	29.49	0.98	10.54	1.77	1.03	4.59
T1_selected_joint_auf_n8192_e3	262,144,000	36.40	1.21	9.59	0.03	2.05	9.80
T1_selected_body128_auf_n8192_e3	48,496,640	31.79	1.06	8.27	0.00	1.83	8.58
T1_selected_fusion_decay7_n8192_e3	52,428,800	27.68	0.92	8.51	0.09	1.19	5.47
T1_selected_ba28_decay7_n8192_e3	931,840	28.12	0.94	7.05	1.83	0.90	4.52
T1_selected_ba56_decay7_n8192_e3	1,863,680	24.61	0.82	7.00	1.79	0.92	4.53
T1_selected_ba128_decay7_n8192_e3	4,259,840	24.06	0.80	7.52	1.80	0.87	4.59
T1_selected_joint_decay7_n8192_e3	262,144,000	33.29	1.11	5.74	0.02	2.17	9.83
T1_selected_body128_decay7_n8192_e3	48,496,640	31.78	1.06	10.84	0.00	1.84	8.58
T1_sweep_auf_n8192_a32_e3	52,428,800	31.64	1.05	8.53	0.02	1.40	5.53
T1_sweep_decay7_n8192_a32_e3	52,428,800	28.24	0.94	16.57	0.02	1.57	5.52
T1_mse50_n8192_e3_lr0.001	52,428,800	17.79	0.59	6.58	0.02	1.39	3.89
T1_mse100_n8192_e3_lr0.001	52,428,800	53.90	1.80	3.45	0.02	1.30	5.53

E WARM REFINEMENT: FINAL ENDPOINTS AND ADDITIONAL COMPLETE COHORTS

The four principal warm refinements start from the same 8k/MSE50 epoch-three checkpoint. “Seen 4k/8k” means the first 4,096/8,192 parent records; “unseen” means the next disjoint 4,096/8,192. They train three AUF epochs at LR 10^{-4} with a fresh optimizer, 32 anchors, seed 42 and global batch eight. Fitting time includes the MSE parent once. Native comparisons are made independently for each workload; no selection of one checkpoint per workload is presented as a single general solution.

Table 12: Warm AUF: does a good starting interface solve everything?

Task	AUF data	$\times$ native	$\times$ MSE parent	Mean TPS	Latency reduction / native	τ	Total fit min
chat	unseen 4k	0.8548 $\times$	0.9624 $\times$	76.51	-17.33%	2.7495	32.39
chat	seen 8k	0.8446 $\times$	0.9509 $\times$	75.59	-19.06%	2.7038	45.32
chat	unseen 8k	0.8431 $\times$	0.9492 $\times$	75.46	-19.32%	2.7030	47.42
chat	seen 4k	0.8420 $\times$	0.9480 $\times$	75.36	-18.58%	2.7403	31.52
code	unseen 4k	0.7417 $\times$	0.8913 $\times$	112.46	-26.52%	4.3499	32.39
code	seen 4k	0.7273 $\times$	0.8740 $\times$	110.28	-27.86%	4.2769	31.52
code	unseen 8k	0.7051 $\times$	0.8473 $\times$	106.92	-31.27%	4.1818	47.42
code	seen 8k	0.7043 $\times$	0.8463 $\times$	106.78	-32.01%	4.1525	45.32
gsm	unseen 8k	1.0606 $\times$	1.0874 $\times$	188.72	+5.53%	6.6218	47.42
gsm	seen 8k	1.0431 $\times$	1.0696 $\times$	185.62	+3.50%	6.5891	45.32
gsm	unseen 4k	1.0413 $\times$	1.0677 $\times$	185.30	+4.01%	6.5340	32.39

Continued

Task	AUF data	$\times$ native	$\times$ MSE parent	Mean TPS	Latency reduction / native	τ	Total fit min
gsm	seen 4k	1.0339 $\times$	1.0601 $\times$	183.97	+3.40%	6.4951	31.52
math	seen 8k	0.9927 $\times$	0.9941 $\times$	218.18	-1.32%	7.8662	45.32
math	unseen 8k	0.9911 $\times$	0.9925 $\times$	217.82	-1.83%	7.8410	47.42
math	seen 4k	0.9717 $\times$	0.9731 $\times$	213.57	-3.20%	7.8145	31.52
math	unseen 4k	0.9694 $\times$	0.9707 $\times$	213.05	-3.42%	7.7785	32.39

The additional early trajectories below provide every completed 128-prompt observation from that refinement inventory. Epoch-one/two checkpoints belong to a three-epoch schedule, not separately optimized shorter schedules. Some trajectories lack complete downstream evaluation; missing observations are not zeros. Row IDs connect the split tables. In the “TPS change” column, the historical denominator is native, not the MSE parent; the separately named parent ratio is the parent comparison. The reported fit cost is cumulative through the checkpoint.

Table 13: Completed early AUF refinement cohorts: performance. These are separate trajectories from the four final warm refinements; epoch costs are cumulative.

ID	Workload	MSE coverage / examples	AUF data / examples	AUF epoch	vs native	Latency reduction	Avg τ	TPS
R1	chat	50% / 8192	unseen (0% overlap) / 4096	2	0.8440 $\times$	-18.25%	2.7174	75.54
R2	chat	50% / 4096	unseen (0% overlap) / 4096	3	0.8321 $\times$	-21.32%	2.6440	74.47
R3	chat	50% / 4096	unseen (0% overlap) / 8192	3	0.8266 $\times$	-21.56%	2.6456	73.99
R4	chat	50% / 8192	unseen (0% overlap) / 4096	1	0.8230 $\times$	-20.95%	2.6693	73.66
R5	chat	50% / 4096	unseen (0% overlap) / 4096	2	0.8177 $\times$	-23.22%	2.6104	73.19
R6	chat	50% / 4096	unseen (0% overlap) / 8192	2	0.8151 $\times$	-23.18%	2.6204	72.96
R7	chat	50% / 4096	unseen (0% overlap) / 4096	1	0.7971 $\times$	-26.00%	2.5586	71.35
R8	chat	100% / 4096	unseen (0% overlap) / 4096	1	0.7961 $\times$	-26.15%	2.5617	71.25
R9	chat	50% / 4096	unseen (0% overlap) / 8192	1	0.7831 $\times$	-27.49%	2.5271	70.09
R10	code	50% / 8192	unseen (0% overlap) / 4096	3	0.7417 $\times$	-26.52%	4.3499	112.46
R11	code	50% / 8192	unseen (0% overlap) / 4096	2	0.7144 $\times$	-29.98%	4.2266	108.33
R12	code	50% / 4096	unseen (0% overlap) / 4096	3	0.6915 $\times$	-35.07%	4.0563	104.86
R13	code	50% / 8192	unseen (0% overlap) / 4096	1	0.6913 $\times$	-35.15%	4.0373	104.81
R14	code	50% / 4096	unseen (0% overlap) / 8192	3	0.6865 $\times$	-34.37%	4.0875	104.09
R15	code	50% / 4096	unseen (0% overlap) / 8192	2	0.6806 $\times$	-36.08%	4.0086	103.20
R16	code	50% / 4096	unseen (0% overlap) / 4096	2	0.6751 $\times$	-37.69%	3.9776	102.35
R17	code	50% / 4096	unseen (0% overlap) / 4096	1	0.6406 $\times$	-45.21%	3.7637	97.13
R18	code	50% / 4096	unseen (0% overlap) / 8192	1	0.6214 $\times$	-48.19%	3.6992	94.22
R19	gsm	50% / 4096	unseen (0% overlap) / 8192	3	1.0445 $\times$	+4.18%	6.6206	185.86
R20	gsm	50% / 4096	unseen (0% overlap) / 4096	3	1.0425 $\times$	+3.68%	6.5104	185.51
R21	gsm	50% / 4096	unseen (0% overlap) / 8192	2	1.0378 $\times$	+3.50%	6.5128	184.66
R22	gsm	50% / 4096	unseen (0% overlap) / 4096	2	1.0056 $\times$	+0.46%	6.4061	178.94
R23	gsm	50% / 8192	unseen (0% overlap) / 4096	2	0.9769 $\times$	-1.74%	6.3272	173.84
R24	gsm	50% / 4096	unseen (0% overlap) / 8192	1	0.9480 $\times$	-4.94%	6.0639	168.70

Continued

ID	Workload	MSE coverage / examples	AUF data / examples	AUF epoch	vs native	Latency reduction	Avg τ	TPS
R25	gsm	50% / 8192	unseen (0% overlap) / 4096	1	0.9474 $\times$	-5.28%	5.9575	168.58
R26	gsm	50% / 4096	unseen (0% overlap) / 4096	1	0.9457 $\times$	-5.61%	5.9792	168.29
R27	math	50% / 4096	unseen (0% overlap) / 8192	3	0.9961 $\times$	-0.79%	7.8768	218.93
R28	math	50% / 4096	unseen (0% overlap) / 4096	3	0.9818 $\times$	-2.53%	7.7689	215.78
R29	math	50% / 8192	unseen (0% overlap) / 4096	3	0.9694 $\times$	-3.42%	7.7785	213.05
R30	math	50% / 4096	unseen (0% overlap) / 8192	2	0.9623 $\times$	-4.03%	7.7204	211.50
R31	math	50% / 8192	unseen (0% overlap) / 4096	2	0.9604 $\times$	-4.66%	7.5780	211.07
R32	math	50% / 4096	unseen (0% overlap) / 4096	2	0.9593 $\times$	-5.01%	7.5572	210.85
R33	math	50% / 4096	unseen (0% overlap) / 8192	1	0.9158 $\times$	-9.72%	7.2421	201.28
R34	math	50% / 8192	unseen (0% overlap) / 4096	1	0.8951 $\times$	-12.01%	7.1195	196.74
R35	math	50% / 4096	unseen (0% overlap) / 4096	1	0.8922 $\times$	-12.15%	7.1537	196.10

Table 14: Completed early AUF refinement cohorts: comparisons. These are separate trajectories from the four final warm refinements; epoch costs are cumulative.

ID	Workload	AUF epoch	MSE parent vs native	vs MSE parent	TPS change
R1	chat	2	0.8882 $\times$	0.9503 $\times$	-15.60%
R2	chat	3	0.8096 $\times$	1.0277 $\times$	-16.79%
R3	chat	3	0.8096 $\times$	1.0210 $\times$	-17.34%
R4	chat	1	0.8882 $\times$	0.9266 $\times$	-17.70%
R5	chat	2	0.8096 $\times$	1.0100 $\times$	-18.23%
R6	chat	2	0.8096 $\times$	1.0068 $\times$	-18.49%
R7	chat	1	0.8096 $\times$	0.9846 $\times$	-20.29%
R8	chat	1	0.8537 $\times$	0.9325 $\times$	-20.39%
R9	chat	1	0.8096 $\times$	0.9672 $\times$	-21.69%
R10	code	3	0.8322 $\times$	0.8913 $\times$	-25.83%
R11	code	2	0.8322 $\times$	0.8585 $\times$	-28.56%
R12	code	3	0.7522 $\times$	0.9194 $\times$	-30.85%
R13	code	1	0.8322 $\times$	0.8307 $\times$	-30.87%
R14	code	3	0.7522 $\times$	0.9126 $\times$	-31.35%
R15	code	2	0.7522 $\times$	0.9048 $\times$	-31.94%
R16	code	2	0.7522 $\times$	0.8974 $\times$	-32.49%
R17	code	1	0.7522 $\times$	0.8516 $\times$	-35.94%
R18	code	1	0.7522 $\times$	0.8261 $\times$	-37.86%
R19	gsm	3	0.9422 $\times$	1.1086 $\times$	+4.45%
R20	gsm	3	0.9422 $\times$	1.1065 $\times$	+4.25%
R21	gsm	2	0.9422 $\times$	1.1015 $\times$	+3.78%
R22	gsm	2	0.9422 $\times$	1.0673 $\times$	+0.56%
R23	gsm	2	0.9753 $\times$	1.0017 $\times$	-2.31%
R24	gsm	1	0.9422 $\times$	1.0063 $\times$	-5.20%
R25	gsm	1	0.9753 $\times$	0.9714 $\times$	-5.26%
R26	gsm	1	0.9422 $\times$	1.0038 $\times$	-5.43%
R27	math	3	0.9764 $\times$	1.0201 $\times$	-0.39%
R28	math	3	0.9764 $\times$	1.0055 $\times$	-1.82%
R29	math	3	0.9986 $\times$	0.9707 $\times$	-3.06%
R30	math	2	0.9764 $\times$	0.9855 $\times$	-3.77%
R31	math	2	0.9986 $\times$	0.9617 $\times$	-3.96%
R32	math	2	0.9764 $\times$	0.9825 $\times$	-4.07%
R33	math	1	0.9764 $\times$	0.9379 $\times$	-8.42%
R34	math	1	0.9986 $\times$	0.8964 $\times$	-10.49%
R35	math	1	0.9764 $\times$	0.9138 $\times$	-10.78%

Table 15: Completed early AUF refinement cohorts: cost. These are separate trajectories from the four final warm refinements; epoch costs are cumulative.

ID	Workload	MSE coverage / examples	AUF data / examples	AUF epoch	MSE min	AUF min	MSE + AUF fit GPU-h
R1	chat	50% / 8192	unseen (0% overlap) / 4096	2	17.79	9.71	0.917
R2	chat	50% / 4096	unseen (0% overlap) / 4096	3	9.56	13.12	0.756
R3	chat	50% / 4096	unseen (0% overlap) / 8192	3	9.56	28.24	1.260
R4	chat	50% / 8192	unseen (0% overlap) / 4096	1	17.79	4.87	0.756
R5	chat	50% / 4096	unseen (0% overlap) / 4096	2	9.56	8.79	0.612
R6	chat	50% / 4096	unseen (0% overlap) / 8192	2	9.56	18.86	0.947
R7	chat	50% / 4096	unseen (0% overlap) / 4096	1	9.56	4.47	0.468
R8	chat	100% / 4096	unseen (0% overlap) / 4096	1	18.30	4.58	0.763
R9	chat	50% / 4096	unseen (0% overlap) / 8192	1	9.56	9.42	0.633
R10	code	50% / 8192	unseen (0% overlap) / 4096	3	17.79	14.60	1.080
R11	code	50% / 8192	unseen (0% overlap) / 4096	2	17.79	9.71	0.917
R12	code	50% / 4096	unseen (0% overlap) / 4096	3	9.56	13.12	0.756
R13	code	50% / 8192	unseen (0% overlap) / 4096	1	17.79	4.87	0.756
R14	code	50% / 4096	unseen (0% overlap) / 8192	3	9.56	28.24	1.260
R15	code	50% / 4096	unseen (0% overlap) / 8192	2	9.56	18.86	0.947
R16	code	50% / 4096	unseen (0% overlap) / 4096	2	9.56	8.79	0.612
R17	code	50% / 4096	unseen (0% overlap) / 4096	1	9.56	4.47	0.468
R18	code	50% / 4096	unseen (0% overlap) / 8192	1	9.56	9.42	0.633
R19	gsm	50% / 4096	unseen (0% overlap) / 8192	3	9.56	28.24	1.260
R20	gsm	50% / 4096	unseen (0% overlap) / 4096	3	9.56	13.12	0.756
R21	gsm	50% / 4096	unseen (0% overlap) / 8192	2	9.56	18.86	0.947
R22	gsm	50% / 4096	unseen (0% overlap) / 4096	2	9.56	8.79	0.612
R23	gsm	50% / 8192	unseen (0% overlap) / 4096	2	17.79	9.71	0.917
R24	gsm	50% / 4096	unseen (0% overlap) / 8192	1	9.56	9.42	0.633
R25	gsm	50% / 8192	unseen (0% overlap) / 4096	1	17.79	4.87	0.756
R26	gsm	50% / 4096	unseen (0% overlap) / 4096	1	9.56	4.47	0.468
R27	math	50% / 4096	unseen (0% overlap) / 8192	3	9.56	28.24	1.260
R28	math	50% / 4096	unseen (0% overlap) / 4096	3	9.56	13.12	0.756
R29	math	50% / 8192	unseen (0% overlap) / 4096	3	17.79	14.60	1.080
R30	math	50% / 4096	unseen (0% overlap) / 8192	2	9.56	18.86	0.947
R31	math	50% / 8192	unseen (0% overlap) / 4096	2	17.79	9.71	0.917
R32	math	50% / 4096	unseen (0% overlap) / 4096	2	9.56	8.79	0.612
R33	math	50% / 4096	unseen (0% overlap) / 8192	1	9.56	9.42	0.633
R34	math	50% / 8192	unseen (0% overlap) / 4096	1	17.79	4.87	0.756

R35	math	50% / 4096	unseen (0% overlap) / 4096	1	9.56	4.47	0.468
-----	------	------------	----------------------------	---	------	------	-------

F COMPLETE MECHANISTIC EVIDENCE

F.1 LIVE ENDPOINTS AND FEATURE RECONSTRUCTION

The primary diagnostic parents are 8k/three-epoch Xavier MSE50 (LR 10^{-3}), AUF and decay CE (LR 10^{-4}). Feature geometry uses 64 training examples and 7,090 sampled positions: 2,994 prompt and 4,096 response. The tables below report live throughput/acceptance separately from reconstruction error and layer cosine. Slot 0–4 is a layer-slot index, not a token position. Lower normalized reconstruction error is favorable, but MSE directly optimizes it.

Table 16: First check: is the MSE advantage real in decoding?

Task	Recipe	Mean TPS	Pooled TPS	$\times$ native	Latency reduction	τ
math	MSE	219.48	216.72	0.9986 $\times$	+0.32%	7.9776
math	Decay CE	194.69	187.71	0.8858 $\times$	-15.09%	6.9474
math	AUF	194.22	186.76	0.8837 $\times$	-15.67%	6.8923
gsm	MSE	173.55	170.79	0.9753 $\times$	-1.80%	6.2568
gsm	Decay CE	163.47	158.50	0.9187 $\times$	-9.70%	5.6926
gsm	AUF	162.05	157.39	0.9107 $\times$	-10.46%	5.6773
code	MSE	126.18	124.01	0.8322 $\times$	-16.54%	4.6912
code	Decay CE	61.56	64.80	0.4060 $\times$	-123.03%	2.4489
code	AUF	58.37	61.90	0.3850 $\times$	-133.47%	2.3456
chat	MSE	79.49	76.54	0.8882 $\times$	-14.51%	2.8125
chat	Decay CE	51.31	49.11	0.5733 $\times$	-78.46%	1.8115
chat	AUF	49.93	47.84	0.5578 $\times$	-83.19%	1.7667

Table 17: Does MSE actually restore the expected features?

Objective	Prompt context error	Response context error
mse	0.216170	0.136147
auf	1.228544	0.940413
decay7	1.189191	0.899731

Table 18: Does MSE actually restore the expected features?

Objective	Layer	Segment	Feature error	Source cosine
mse	0	prompt	0.060359	0.975595
mse	0	response	0.033437	0.984793
mse	1	prompt	0.191632	0.899016
mse	1	response	0.068966	0.965129
mse	2	prompt	0.192591	0.901045
mse	2	response	0.083779	0.957394
mse	3	prompt	0.191474	0.902611
mse	3	response	0.110141	0.944060
mse	4	prompt	0.180501	0.909023
mse	4	response	0.090359	0.954783
auf	0	prompt	2.742802	0.195425
auf	0	response	2.794326	0.214868
auf	1	prompt	2.859036	0.112855
auf	1	response	2.540209	0.120631
auf	2	prompt	21.592222	0.141729
auf	2	response	12.066430	0.175011
auf	3	prompt	6.882195	0.144134
auf	3	response	4.716092	0.226237
auf	4	prompt	3.814032	0.190191
auf	4	response	3.408715	0.266906
decay7	0	prompt	2.710681	0.201154

Continued

Objective	Layer	Segment	Feature error	Source cosine
decay7	0	response	2.753600	0.220411
decay7	1	prompt	2.827380	0.119284
decay7	1	response	2.522013	0.126873
decay7	2	prompt	21.544376	0.145896
decay7	2	response	12.034999	0.179880
decay7	3	prompt	6.830830	0.151039
decay7	3	response	4.675968	0.234286
decay7	4	prompt	3.730459	0.201127
decay7	4	response	3.298069	0.278744

F.2 HELD-OUT, COMMON-INPUT COMPARISONS

Eight prompt-length quantiles per workload give 32 held-out examples, with eight fixed anchors per example (64 blocks per task). Each variant sees the same saved native continuation. Full CE averages all valid labels irrespective of training objective. AUF support is the fraction of labels that would receive AUF loss, including the first wrong label; it is not token accuracy. Correct prefix excludes the verifier bonus. Prompt/response context errors average positions within each record, then records equally. Midpoints average all five maps without fitting. These small probes are not a new final test set.

Table 19: MATH

Variant	Full CE ↓	AUF support	Correct prefix ↑	Prompt error ↓	Response error ↓
MSE	1.6939	51.81%	6.7656	0.1821	0.1236
Decay halfway to MSE	1.8043	49.79%	6.4844	0.7590	0.4928
Decay CE	1.9534	47.13%	6.0625	1.3852	0.9058
Warm unseen 4k	2.0634	52.45%	6.9219	0.3910	0.3867
Warm seen 4k	2.0695	50.64%	6.6094	0.3947	0.3827
AUF halfway to MSE	2.1282	48.51%	6.2812	0.7768	0.5110
Warm unseen 8k	2.1507	54.47%	7.2188	0.4347	0.4084
Warm seen 8k	2.1760	56.49%	7.5625	0.4362	0.4088
AUF	2.6028	45.74%	5.8438	1.4131	0.9333
Random initial	8.6261	6.81%	0.0000	4.4041	4.9689

Table 20: GSM

Variant	Full CE ↓	AUF support	Correct prefix ↑	Prompt error ↓	Response error ↓
Decay halfway to MSE	1.5407	55.01%	7.1875	0.7967	0.4933
Decay CE	1.6564	50.70%	6.5312	1.4673	0.8850
Warm seen 4k	1.7356	50.91%	6.5625	0.4074	0.3686
Warm seen 8k	1.7586	57.48%	7.5781	0.4506	0.3911
Warm unseen 8k	1.7793	55.65%	7.3125	0.4535	0.3911
MSE	1.7817	53.07%	6.8750	0.2026	0.1146
Warm unseen 4k	1.8745	55.65%	7.2812	0.4053	0.3692
AUF halfway to MSE	1.9572	55.76%	7.3125	0.8212	0.5098
AUF	2.0566	54.14%	7.0781	1.5231	0.9074
Random initial	9.6898	7.00%	0.0156	4.0335	4.6817

Table 21: CODE

Variant	Full CE ↓	AUF support	Correct prefix ↑	Prompt error ↓	Response error ↓
MSE	2.6039	40.46%	4.8750	0.3942	0.2864
Warm unseen 4k	3.4776	32.24%	3.6719	0.5446	0.4510
Warm unseen 8k	3.5000	32.79%	3.7344	0.5785	0.4877
Warm seen 8k	3.5864	32.02%	3.6250	0.5754	0.4818
Warm seen 4k	3.6100	33.77%	3.8906	0.5456	0.4543
Decay halfway to MSE	3.7045	25.33%	2.6406	0.9133	0.6970
AUF halfway to MSE	4.1200	28.62%	3.1250	0.9301	0.7159
Decay CE	4.6468	17.98%	1.5625	1.5436	1.2464

Continued

Variant	Full CE ↓	AUF support	Correct prefix ↑	Prompt error ↓	Response error ↓
AUF	5.5081	17.00%	1.4219	1.5820	1.2868
Random initial	9.8854	7.24%	0.0312	3.7889	4.6308

Table 22: CHAT

Variant	Full CE ↓	AUF support	Correct prefix ↑	Prompt error ↓	Response error ↓
MSE	3.8074	28.50%	2.9062	0.3017	0.3255
Warm unseen 8k	4.1924	26.64%	2.6406	0.4872	0.4895
Warm seen 4k	4.2015	26.99%	2.6719	0.4507	0.4547
Warm seen 8k	4.2630	24.65%	2.3594	0.4868	0.4873
Warm unseen 4k	4.2904	24.42%	2.3281	0.4504	0.4519
Decay halfway to MSE	4.5984	21.03%	1.8750	0.9437	0.7632
AUF halfway to MSE	4.9769	22.78%	2.1094	0.9668	0.7841
Decay CE	5.6398	16.36%	1.2188	1.7617	1.3749
AUF	6.2094	13.55%	0.8438	1.8059	1.4210
Random initial	10.7415	7.48%	0.0000	4.9159	4.0375

F.3 ALL FOUR LIVE MIDPOINT INTERVENTIONS

The reference for parent-relative latency reduction is the named cold AUF/decay endpoint. All four midpoint evaluations contain 128 requests. Timing passes are separate. The interpolation changes all five maps, not one isolated feature statistic.

Table 23: Intervention: move the token-trained maps toward MSE

Task	Compared with	Variant	Mean TPS	Pooled TPS	× native	× cold parent	Total seconds	Latency reduction / parent	τ
code	AUF	MSE	126.18	124.01	0.8322×	2.1618×	569.96	+50.08%	4.6912
code	AUF	AUF	81.64	85.41	0.5384×	1.3986×	827.50	+27.53%	3.2512
		halfway to MSE							
code	AUF	AUF	58.37	61.90	0.3850×	1.0000×	1141.85	+0.00%	2.3456
code	Decay CE	MSE	126.18	124.01	0.8322×	2.0498×	569.96	+47.75%	4.6912
code	Decay CE	Decay CE	84.68	88.03	0.5585×	1.3756×	802.88	+26.39%	3.3368
		halfway to MSE							
code	Decay CE	Decay CE	61.56	64.80	0.4060×	1.0000×	1090.76	+0.00%	2.4489
math	AUF	MSE	219.48	216.72	0.9986×	1.1300×	406.20	+13.83%	7.9776
math	AUF	AUF	206.81	201.87	0.9410×	1.0648×	436.09	+7.48%	7.5624
		halfway to MSE							
math	AUF	AUF	194.22	186.76	0.8837×	1.0000×	471.37	+0.00%	6.8923
math	Decay CE	MSE	219.48	216.72	0.9986×	1.1273×	406.20	+13.39%	7.9776
math	Decay CE	Decay CE	208.80	203.73	0.9500×	1.0725×	432.11	+7.86%	7.5748
		halfway to MSE							
math	Decay CE	Decay CE	194.69	187.71	0.8858×	1.0000×	468.98	+0.00%	6.9474

F.4 AUF BRANCH, RANK AND DIRECTIONAL INTERVENTIONS

Let $\Delta W_i = W_i - W_{0,i}$. Restoring slot i replaces W_i by $W_{0,i}$. A rank- k intervention truncates ΔW_i by SVD and adds it back to the full random $W_{0,i}$. Fusion-sensitive removal takes 32 leading right-singular directions Q_i of $F_i \Delta W_i$ and uses $W_{0,i} + \Delta W_i(I - Q_i Q_i^\top)$. Random removal uses 32 random orthonormal directions per slot: a rank-matched, not energy-matched control.

For $z_0 = \sum_i F_i W_{0,i} h_i$ and $\delta = \sum_i F_i \Delta W_i h_i$, define

$$\delta_{\parallel} = \frac{\delta^\top z_0}{\max(\|z_0\|^2, 10^{-12})} z_0, \quad \delta_{\perp} = \delta - \delta_{\parallel}. \quad (15)$$

Radial/perpendicular interventions feed $N(z_0 + \delta_{\parallel})$ or $N(z_0 + \delta_{\perp})$; the matched hook control reconstructs $N(z_0 + \delta)$. In the table, parent denotes the unmodified AUF checkpoint. Nodes denote separate L40S measurements, not distinct training seeds. The unmodified Code parent completed at 58.5547 TPS, 0.3862 times native, 1,138.60 summed seconds, and acceptance 2.3456. Code sensitive/random-removal cohorts were incomplete; there is no complete cross-domain replication of those two destructive interventions.

Table 24: Full Math intervention results

Intervention	Node	Control	Mean TPS	$\times$ native (TPS)	Summed request seconds	τ	TPS $\times$ control	Latency speedup $\times$ control	Latency reduction
parent	node08	parent	195.3339	0.8887	469.13	6.8923	1.0000	1.0000	0.00%
parent	node09	parent	194.7817	0.8862	470.59	6.8923	1.0000	1.0000	0.00%
random_remove	node09	parent	194.4424	0.8847	470.94	6.8735	0.9983	0.9993	-0.07%
hook_control	node08	hook_control	193.6967	0.8813	472.79	6.8956	1.0000	1.0000	0.00%
hook_control	node09	hook_control	193.4648	0.8802	473.26	6.8956	1.0000	1.0000	0.00%
restore1	node09	parent	189.0736	0.8603	482.30	6.7770	0.9707	0.9757	-2.49%
restore0	node08	parent	188.0331	0.8555	488.25	6.6363	0.9626	0.9608	-4.07%
restore2	node08	parent	180.0708	0.8193	511.39	6.3676	0.9219	0.9174	-9.01%
perpendicular	node09	hook_control	155.3280	0.7067	601.45	5.4175	0.8029	0.7869	-27.09%
delta128	node09	parent	153.9099	0.7003	601.05	5.4182	0.7902	0.7829	-27.72%
restore3	node09	parent	100.1638	0.4557	905.34	3.5735	0.5142	0.5198	-92.39%
restore4	node08	parent	71.1581	0.3238	1386.93	2.3407	0.3643	0.3383	-195.64%
delta32	node08	parent	58.2957	0.2652	1542.19	2.1151	0.2984	0.3042	-228.73%
fusion_remove	node08	parent	38.6362	0.1758	2288.12	1.4086	0.1978	0.2050	-387.73%
delta2	node09	parent	29.2849	0.1332	3067.04	1.0780	0.1503	0.1534	-551.75%
delta1	node08	parent	29.0863	0.1323	3112.75	1.0377	0.1489	0.1507	-563.51%
radial	node08	hook_control	28.2457	0.1285	3185.37	1.0155	0.1458	0.1484	-573.73%

F.5 ALL RECORDED ENERGY SPECTRA

For each objective and slot, matrices W_i , ΔW_i and $F_i \Delta W_i$ have distinct spectra. Correction activations are $F_i \Delta W_i h_i$, evaluated separately on prompt and response positions and not centered. Each complete-spectrum cell lists rank 1/2/32/128 percentages in that order. Energy concentration is not substituted for a live speed measurement. In particular, none of the MSE energy rows establishes a successful compressed MSE deployment.

Table 25: Energy measurements: what they mean, and what they do not

Objective	Layer	Top-2 map-update energy	Top-2 fused-update energy	Top-2 prompt correction energy	Top-2 response correction energy
mse	0	2.52%	30.19%	76.19%	58.65%
mse	1	0.81%	5.17%	99.96%	55.79%
mse	2	1.04%	2.38%	99.79%	84.34%
mse	3	0.73%	2.42%	98.96%	58.04%
mse	4	0.71%	3.75%	93.50%	67.39%
auf	0	11.62%	48.45%	42.54%	43.73%
auf	1	5.12%	22.07%	99.38%	51.03%
auf	2	4.46%	9.23%	97.91%	51.68%
auf	3	3.16%	10.75%	84.13%	48.91%
auf	4	3.63%	16.51%	91.39%	84.28%
decay7	0	10.38%	47.21%	42.56%	43.27%
decay7	1	4.51%	20.11%	99.20%	52.15%
decay7	2	4.18%	8.74%	97.27%	50.14%
decay7	3	3.09%	10.83%	83.44%	47.88%
decay7	4	3.41%	15.95%	91.70%	84.10%

1350

1351

Table 26: Complete recorded spectra

Objective	Slot	Quantity	Segment	Energy % at ranks 1 / 2 / 32 / 128
auf	0	W_matrix_energy	not applicable	0.60 / 1.17 / 7.99 / 18.46
auf	0	activation_ correction_energy	prompt	27.20 / 42.54 / 97.22 / 99.90
auf	0	activation_ correction_energy	response	24.78 / 43.73 / 97.01 / 99.89
auf	0	delta_matrix_energy	not applicable	6.07 / 11.62 / 65.11 / 91.55
auf	0	fused_delta_matrix_ energy	not applicable	27.65 / 48.45 / 95.18 / 99.44
auf	1	W_matrix_energy	not applicable	0.23 / 0.45 / 4.79 / 15.42
auf	1	activation_ correction_energy	prompt	98.88 / 99.38 / 99.87 / 99.99
auf	1	activation_ correction_energy	response	44.43 / 51.03 / 88.57 / 98.78
auf	1	delta_matrix_energy	not applicable	2.61 / 5.12 / 39.31 / 74.99
auf	1	fused_delta_matrix_ energy	not applicable	11.30 / 22.07 / 69.40 / 91.43
auf	2	W_matrix_energy	not applicable	0.20 / 0.40 / 4.72 / 15.38
auf	2	activation_ correction_energy	prompt	96.96 / 97.91 / 99.55 / 99.96
auf	2	activation_ correction_energy	response	45.33 / 51.68 / 86.62 / 98.92
auf	2	delta_matrix_energy	not applicable	2.24 / 4.46 / 39.08 / 78.04
auf	2	fused_delta_matrix_ energy	not applicable	5.19 / 9.23 / 59.83 / 92.77
auf	3	W_matrix_energy	not applicable	0.16 / 0.31 / 4.13 / 14.67
auf	3	activation_ correction_energy	prompt	78.41 / 84.13 / 94.49 / 99.01
auf	3	activation_ correction_energy	response	44.36 / 48.91 / 80.41 / 96.90
auf	3	delta_matrix_energy	not applicable	1.69 / 3.16 / 29.35 / 68.51
auf	3	fused_delta_matrix_ energy	not applicable	6.27 / 10.75 / 51.55 / 85.96
auf	4	W_matrix_energy	not applicable	0.21 / 0.36 / 4.04 / 14.49
auf	4	activation_ correction_energy	prompt	75.94 / 91.39 / 97.03 / 99.22
auf	4	activation_ correction_energy	response	82.31 / 84.28 / 94.56 / 98.73
auf	4	delta_matrix_energy	not applicable	2.38 / 3.63 / 23.66 / 57.51
auf	4	fused_delta_matrix_ energy	not applicable	12.13 / 16.51 / 37.93 / 68.53
decay7	0	W_matrix_energy	not applicable	0.55 / 1.08 / 7.82 / 18.41
decay7	0	activation_ correction_energy	prompt	27.52 / 42.56 / 96.80 / 99.87
decay7	0	activation_ correction_energy	response	24.71 / 43.27 / 96.58 / 99.87
decay7	0	delta_matrix_energy	not applicable	5.38 / 10.38 / 61.85 / 90.42
decay7	0	fused_delta_matrix_ energy	not applicable	27.17 / 47.21 / 94.31 / 99.31
decay7	1	W_matrix_energy	not applicable	0.23 / 0.44 / 4.75 / 15.41
decay7	1	activation_ correction_energy	prompt	98.65 / 99.20 / 99.83 / 99.98
decay7	1	activation_ correction_energy	response	45.51 / 52.15 / 87.90 / 98.60
decay7	1	delta_matrix_energy	not applicable	2.33 / 4.51 / 36.47 / 72.57
decay7	1	fused_delta_matrix_ energy	not applicable	10.35 / 20.11 / 66.59 / 90.07

1403

1404	<i>Continued</i>				
1405	Objective	Slot	Quantity	Segment	Energy % at ranks 1 / 2 / 32 / 128
1406					
1407	decay7	2	W_matrix_energy	not applicable	0.21 / 0.40 / 4.75 / 15.46
1408	decay7	2	activation_ correction_energy	prompt	96.22 / 97.27 / 99.42 / 99.94
1409	decay7	2	activation_ correction_energy	response	43.81 / 50.14 / 85.73 / 98.74
1410	decay7	2	delta_matrix_energy	not applicable	2.15 / 4.18 / 37.31 / 76.74
1411	decay7	2	fused_delta_matrix_ energy	not applicable	4.66 / 8.74 / 58.60 / 92.27
1412	decay7	3	W_matrix_energy	not applicable	0.16 / 0.32 / 4.15 / 14.71
1413	decay7	3	activation_ correction_energy	prompt	78.20 / 83.44 / 94.03 / 98.88
1414	decay7	3	activation_ correction_energy	response	43.29 / 47.88 / 79.25 / 96.57
1415	decay7	3	delta_matrix_energy	not applicable	1.70 / 3.09 / 28.76 / 67.87
1416	decay7	3	fused_delta_matrix_ energy	not applicable	6.31 / 10.83 / 51.69 / 86.02
1417	decay7	4	W_matrix_energy	not applicable	0.22 / 0.36 / 4.03 / 14.46
1418	decay7	4	activation_ correction_energy	prompt	76.39 / 91.70 / 96.98 / 99.16
1419	decay7	4	activation_ correction_energy	response	82.12 / 84.10 / 94.19 / 98.55
1420	decay7	4	delta_matrix_energy	not applicable	2.27 / 3.41 / 21.76 / 54.77
1421	decay7	4	fused_delta_matrix_ energy	not applicable	11.99 / 15.95 / 36.06 / 66.32
1422	mse	0	W_matrix_energy	not applicable	3.60 / 5.68 / 12.97 / 26.35
1423	mse	0	activation_ correction_energy	prompt	50.63 / 76.19 / 92.43 / 97.83
1424	mse	0	activation_ correction_energy	response	52.06 / 58.65 / 85.34 / 95.59
1425	mse	0	delta_matrix_energy	not applicable	1.59 / 2.52 / 7.16 / 18.24
1426	mse	0	fused_delta_matrix_ energy	not applicable	22.80 / 30.19 / 60.20 / 77.17
1427	mse	1	W_matrix_energy	not applicable	1.59 / 2.54 / 9.15 / 22.82
1428	mse	1	activation_ correction_energy	prompt	99.94 / 99.96 / 99.98 / 99.99
1429	mse	1	activation_ correction_energy	response	52.61 / 55.79 / 75.67 / 89.65
1430	mse	1	delta_matrix_energy	not applicable	0.48 / 0.81 / 4.76 / 15.52
1431	mse	1	fused_delta_matrix_ energy	not applicable	2.70 / 5.17 / 23.27 / 46.80
1432	mse	2	W_matrix_energy	not applicable	2.39 / 3.83 / 10.73 / 22.91
1433	mse	2	activation_ correction_energy	prompt	99.68 / 99.79 / 99.94 / 99.98
1434	mse	2	activation_ correction_energy	response	69.37 / 84.34 / 91.60 / 96.67
1435	mse	2	delta_matrix_energy	not applicable	0.63 / 1.04 / 5.05 / 15.99
1436	mse	2	fused_delta_matrix_ energy	not applicable	1.24 / 2.38 / 24.34 / 57.01
1437	mse	3	W_matrix_energy	not applicable	1.63 / 2.46 / 9.05 / 21.15
1438	mse	3	activation_ correction_energy	prompt	98.43 / 98.96 / 99.52 / 99.78
1439	mse	3	activation_ correction_energy	response	48.60 / 58.04 / 76.42 / 89.34
1440	mse	3	delta_matrix_energy	not applicable	0.46 / 0.73 / 4.78 / 16.00
1441	mse	3	fused_delta_matrix_ energy	not applicable	1.27 / 2.42 / 19.60 / 47.95
1442	mse	4	W_matrix_energy	not applicable	2.39 / 3.60 / 9.94 / 21.23
1443	mse	4	W_matrix_energy	not applicable	2.39 / 3.60 / 9.94 / 21.23

Continued

Objective	Slot	Quantity	Segment	Energy % at ranks 1 / 2 / 32 / 128
mse	4	activation_ correction_energy	prompt	88.05 / 93.50 / 96.99 / 98.29
mse	4	activation_ correction_energy	response	64.72 / 67.39 / 80.66 / 89.34
mse	4	delta_matrix_energy	not applicable	0.46 / 0.71 / 4.47 / 15.06
mse	4	fused_delta_matrix_ energy	not applicable	2.35 / 3.75 / 10.83 / 28.20

F.6 RECTANGULAR IDENTITY AND LEARNING RATE

All new maps start at $[I_{2560} \ 0_{2560 \times 1536}]$; all three fits use 8,192 examples, three epochs and LR 10^{-3} . MSE uses quarter coverage, and token objectives use 32 response anchors. Training cap remains 4,096 and evaluation cap 2,048. The old MSE25 comparison has the same LR and differs only in initialization; the token comparisons also change LR by tenfold. Four disjoint GPU shards are combined for each 128-request cohort; each request is still generated serially. All 12 cohorts match saved native outputs on 128/128. There are no repeated timing trials to prove that the GSM8K timing reversal is noise.

Table 27: Does rectangular-identity initialization close the gap?

Dataset	Loss	TPS	Avg. τ length	Speed vs native
Math	MSE25	213.70	7.983	0.9723×
Math	AUF	206.16	7.326	0.9380×
Math	Decay CE	203.74	7.348	0.9270×
GSM8K	AUF	175.35	6.166	0.9854×
GSM8K	Decay CE	169.72	6.073	0.9538×
GSM8K	MSE25	166.70	6.217	0.9368×
Code	MSE25	119.49	4.569	0.7881×
Code	Decay CE	76.02	3.042	0.5014×
Code	AUF	73.10	2.931	0.4821×
Chat	MSE25	76.13	2.747	0.8506×
Chat	Decay CE	61.11	2.183	0.6828×
Chat	AUF	60.77	2.163	0.6790×

Table 28: Old versus identity initialization

Dataset	Loss	Old TPS	Identity TPS	TPS change	Old avg. length	Identity avg. length	Length change
Math	MSE25	218.75	213.70	-2.31%	7.973	7.983	+0.14%
Math	AUF	194.22	206.16	+6.15%	6.892	7.326	+6.29%
Math	Decay CE	194.69	203.74	+4.65%	6.947	7.348	+5.77%
GSM8K	AUF	162.05	175.35	+8.21%	5.677	6.166	+8.60%
GSM8K	Decay CE	163.47	169.72	+3.82%	5.693	6.073	+6.68%
GSM8K	MSE25	172.35	166.70	-3.28%	6.204	6.217	+0.21%
Code	MSE25	125.98	119.49	-5.15%	4.690	4.569	-2.58%
Code	Decay CE	61.56	76.02	+23.49%	2.449	3.042	+24.20%
Code	AUF	58.37	73.10	+25.24%	2.346	2.931	+24.94%
Chat	MSE25	78.31	76.13	-2.78%	2.802	2.747	-1.94%
Chat	Decay CE	51.31	61.11	+19.09%	1.812	2.183	+20.52%
Chat	AUF	49.93	60.77	+21.73%	1.767	2.163	+22.41%

Table 29: Old versus identity initialization

Loss	Fit wall time	Fit GPU-hours (two GPUs)
MSE25	11.11 min	0.370
AUF	30.14 min	1.005
Decay CE	29.54 min	0.985

G ADDITIONAL COMPLETE PHASE 2 OBSERVATIONS

This inventory retains completed earlier one-epoch and continued-training observations separately from the selected three-epoch recipe, rather than pooling them into the headline comparison. A run identifier records pair, architecture, objective, example count, anchor budget and epochs where present. “Plus2” denotes two additional epochs of a continued run, not an independent fresh three-epoch fit. Its fitting time is the recorded continuation duration. “Historical” denotes a separately recorded pass whose physical node is not present in the consolidated row. Neither shared names nor copied files create independent replications. The selected T1–T4 table takes precedence over these earlier recipes.

Table 30: All completed Phase 2 standalone observations (including earlier recipes)

Pair/task	Run or control	Node	Mean TPS	$\times$ native	Descriptive AR ratio	τ	Fit min	Parameters
T1/chat	T1_selected_body128_auf_n8192_e3	node08	91.9628	1.0275	2.6209	3.4139	31.7912	48496640
T1/chat	T1_selected_body128_decay7_n8192_e3	node09	90.2557	1.0084	2.5723	3.3477	31.7757	48496640
T1/chat	native	node08	89.5033	1.0000	2.5508	3.3236	—	—
T1/chat	T1_mse100_n4096_e3_lr0.001	node09	76.4124	0.8537	2.1777	2.6811	18.3032	52428800
T1/chat	T1_priority_auf_n16384_e3_lr1e4	historical	56.3743	0.6299	1.6067	1.9829	59.0507	52428800
T1/chat	T1_sweep_decay7_n8192_a32_e3	node09	51.3131	0.5733	1.4624	1.8115	28.2368	52428800
T1/chat	T1_sweep_auf_n8192_a32_e3	historical	49.9253	0.5578	1.4229	1.7667	31.6433	52428800
T1/chat	T1_sweep_decay7_n4096_a32_e3	historical	43.1691	0.4823	1.2303	1.5760	13.0235	52428800
T1/chat	T1_sweep_auf_n4096_a32_e3	historical	41.9533	0.4687	1.1957	1.5320	13.2874	52428800
T1/chat	T1_priority_auf_n4096_plus2_lr1e4	historical	41.2074	0.4604	1.1744	1.5191	8.9126	52428800
T1/chat	T1_fivew_auf_n4096_a32	historical	35.2203	0.3935	1.0038	1.2948	4.7556	52428800
T1/chat	ar	node08	35.0880	0.3920	1.0000	1.0000	—	—
T1/code	T1_selected_body128_auf_n8192_e3	node08	155.3287	1.0244	4.1424	5.8636	31.7912	48496640
T1/code	T1_selected_body128_decay7_n8192_e3	node09	153.8642	1.0148	4.1034	5.7508	31.7757	48496640
T1/code	native	node08	151.6250	1.0000	4.0437	5.6441	—	—
T1/code	T1_mse100_n8192_e3_lr0.001	node08	125.5835	0.8283	3.3492	4.6968	53.9014	52428800
T1/code	T1_mse100_n4096_e3_lr0.001	node09	118.0456	0.7785	3.1481	4.4341	18.3032	52428800
T1/code	T1_selected_joint_decay7_n8192_e3	node08	70.9838	0.4682	1.8931	2.8271	33.2872	262144000
T1/code	T1_selected_joint_auf_n8192_e3	node08	68.2183	0.4499	1.8193	2.7418	36.3991	262144000
T1/code	T1_priority_auf_n16384_e3_lr1e4	historical	67.6783	0.4464	1.8049	2.7210	59.0507	52428800
T1/code	T1_sweep_decay7_n8192_a32_e3	node09	61.5596	0.4060	1.6417	2.4489	28.2368	52428800
T1/code	T1_sweep_auf_n8192_a32_e3 / selected8k_unmodified	node09	58.5547	0.3862	1.5616	2.3456	31.6433	52428800
T1/code	T1_sweep_auf_n8192_a32_e3	historical	58.3696	0.3850	1.5566	2.3456	31.6433	52428800
T1/code	T1_sweep_decay7_n4096_a32_e3	historical	50.3969	0.3324	1.3440	1.9998	13.0235	52428800
T1/code	T1_priority_decay7_n4096_plus2_lr1e4	historical	49.1700	0.3243	1.3113	1.9418	8.7649	52428800
T1/code	T1_sweep_auf_n4096_a32_e3	historical	48.2183	0.3180	1.2859	1.9104	13.2874	52428800
T1/code	T1_priority_auf_n4096_plus2_lr1e4	historical	47.9888	0.3165	1.2798	1.9023	8.9126	52428800
T1/code	T1_selected_ba128_decay7_n8192_e3	node08	43.8303	0.2891	1.1689	1.7185	24.0569	4259840
T1/code	T1_selected_ba128_auf_n8192_e3	node08	41.5391	0.2740	1.1078	1.6302	29.4859	4259840
T1/code	T1_selected_ba56_decay7_n8192_e3	node09	38.7609	0.2556	1.0337	1.4981	24.6055	1863680
T1/code	ar	node08	37.4970	0.2473	1.0000	1.0000	—	—
T1/code	T1_selected_ba56_auf_n8192_e3	node08	37.3163	0.2461	0.9952	1.4330	25.2175	1863680
T1/code	T1_fivew_auf_n4096_a32	historical	36.7751	0.2425	0.9807	1.4041	4.7556	52428800
T1/code	T1_selected_ba28_decay7_n8192_e3	node09	35.8736	0.2366	0.9567	1.3653	28.1235	931840
T1/code	T1_selected_ba28_auf_n8192_e3	node08	34.7612	0.2293	0.9270	1.3335	29.7632	931840
T1/code	T1_selected_fusion_auf_n8192_e3	node09	33.8244	0.2231	0.9021	1.2953	26.8966	52428800
T1/code	T1_selected_fusion_decay7_n8192_e3	node09	33.6595	0.2220	0.8977	1.2921	27.6807	52428800
T1/gsm	T1_selected_body128_auf_n8192_e3	node08	194.8627	1.0951	5.1603	7.0986	31.7912	48496640

1566

Continued

1567

1568

1569

1570

1571

1572

1573

1574

1575

1576

1577

1578

1579

1580

1581

1582

1583

1584

1585

1586

1587

1588

1589

1590

1591

1592

1593

1594

1595

1596

1597

1598

1599

1600

1601

1602

1603

1604

1605

1606

1607

1608

1609

1610

1611

1612

1613

1614

1615

1616

1617

1618

1619

Pair/task	Run or control	Node	Mean TPS	$\times$ native	Descriptive AR ratio	τ	Fit min	Parameters
T1/gsm	T1_selected_body128_decay7_ n8192_e3	node09	187.6579	1.0546	4.9695	6.8111	31.7757	48496640
T1/gsm	native	node08	177.9411	1.0000	4.7121	6.4788	—	—
T1/gsm	T1_mse100_n8192_e3_lr0.001	node08	176.1834	0.9901	4.6656	6.2244	53.9014	52428800
T1/gsm	T1_priority_auf_n16384_e3_ lr1e4	historical	175.1087	0.9841	4.6371	6.2470	59.0507	52428800
T1/gsm	T1_mse100_n4096_e3_lr0.001	node09	174.3330	0.9797	4.6166	6.1869	18.3032	52428800
T1/gsm	T1_sweep_decay7_n8192_a32_e3	node08	163.4700	0.9187	4.3289	5.6926	28.2368	52428800
T1/gsm	T1_sweep_auf_n8192_a32_e3	historical	162.0495	0.9107	4.2913	5.6773	31.6433	52428800
T1/gsm	T1_sweep_decay7_n4096_a32_e3	historical	134.9602	0.7585	3.5739	4.7496	13.0235	52428800
T1/gsm	T1_priority_decay7_n4096_ plus2_lr1e4	historical	132.1416	0.7426	3.4993	4.6352	8.7649	52428800
T1/gsm	T1_sweep_auf_n4096_a32_e3	historical	131.9362	0.7415	3.4939	4.5797	13.2874	52428800
T1/gsm	T1_priority_auf_n4096_plus2_ lr1e4	historical	125.7897	0.7069	3.3311	4.4761	8.9126	52428800
T1/gsm	T1_fivew_decay7_n4096_a32	historical	78.3291	0.4402	2.0743	2.7460	4.5598	52428800
T1/gsm	T1_fivew_auf_n4096_a32	historical	74.3406	0.4178	1.9686	2.6269	4.7556	52428800
T1/gsm	ar	node08	37.7622	0.2122	1.0000	1.0000	—	—
T1/math	T1_selected_body128_auf_ n8192_e3	node08	230.8243	1.0502	6.1004	8.6097	31.7912	48496640
T1/math	T1_selected_body128_decay7_ n8192_e3	node09	226.1528	1.0290	5.9770	8.4241	31.7757	48496640
T1/math	native	node08	219.7874	1.0000	5.8087	8.2108	—	—
T1/math	T1_mse100_n8192_e3_lr0.001	node08	219.3158	0.9979	5.7963	7.9994	53.9014	52428800
T1/math	T1_mse100_n4096_e3_lr0.001	node09	219.1429	0.9971	5.7917	7.9711	18.3032	52428800
T1/math	T1_priority_auf_n16384_e3_ lr1e4	historical	207.2180	0.9428	5.4766	7.4294	59.0507	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_unmodified	node08	195.3339	0.8887	5.1625	6.8923	31.6433	52428800
T1/math	T1_selected_joint_decay7_ n8192_e3	node08	195.2239	0.8882	5.1596	6.9644	33.2872	262144000
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_unmodified	node09	194.7817	0.8862	5.1479	6.8923	31.6433	52428800
T1/math	T1_sweep_decay7_n8192_a32_e3	node08	194.6915	0.8858	5.1455	6.9474	28.2368	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_random_remove	node09	194.4424	0.8847	5.1389	6.8735	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3	historical	194.2206	0.8837	5.1330	6.8923	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_hook_control	node08	193.6967	0.8813	5.1192	6.8956	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_hook_control	node09	193.4648	0.8802	5.1131	6.8956	31.6433	52428800
T1/math	T1_selected_joint_auf_n8192_e3	node08	193.3218	0.8796	5.1093	6.9562	36.3991	262144000
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_restore1	node09	189.0736	0.8603	4.9970	6.7770	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_restore0	node08	188.0331	0.8555	4.9695	6.6363	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_restore2	node08	180.0708	0.8193	4.7591	6.3676	31.6433	52428800
T1/math	T1_sweep_decay7_n4096_a32_e3	historical	173.8943	0.7912	4.5958	6.1506	13.0235	52428800
T1/math	T1_priority_decay7_n4096_ plus2_lr1e4	historical	171.4343	0.7800	4.5308	6.0574	8.7649	52428800
T1/math	T1_sweep_auf_n4096_a32_e3	historical	170.1361	0.7741	4.4965	6.0168	13.2874	52428800
T1/math	T1_priority_auf_n4096_plus2_ lr1e4	historical	169.2277	0.7700	4.4725	5.9179	8.9126	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_perpendicular	node09	155.3280	0.7067	4.1052	5.4175	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_delta128	node09	153.9099	0.7003	4.0677	5.4182	31.6433	52428800
T1/math	T1_selected_ba128_decay7_ n8192_e3	node08	149.9202	0.6821	3.9622	5.2374	24.0569	4259840
T1/math	T1_selected_ba128_auf_n8192_ e3	node08	143.0149	0.6507	3.7797	4.9728	29.4859	4259840
T1/math	T1_sweep_auf_n4096_a8_e3	historical	127.2140	0.5788	3.3621	4.6098	10.3192	52428800
T1/math	T1_selected_ba56_decay7_ n8192_e3	node09	119.0842	0.5418	3.1473	4.2197	24.6055	1863680
T1/math	T1_selected_ba56_auf_n8192_e3	node08	109.9603	0.5003	2.9061	3.7969	25.2175	1863680
T1/math	T1_fivew_decay7_n4096_a32	historical	107.7487	0.4902	2.8477	3.7605	4.5598	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_restore3	node09	100.1638	0.4557	2.6472	3.5735	31.6433	52428800
T1/math	T1_fivew_auf_n4096_a32	historical	95.3304	0.4337	2.5195	3.4532	4.7556	52428800

Continued

Pair/task	Run or control	Node	Mean TPS	$\times$ native	Descriptive AR ratio	τ	Fit min	Parameters
T1/math	T1_selected_ba28_decay7_n8192_e3	node09	93.2439	0.4242	2.4643	3.2391	28.1235	931840
T1/math	T1_selected_ba28_auf_n8192_e3	node08	82.5736	0.3757	2.1823	2.8760	29.7632	931840
T1/math	T1_selected_fusion_decay7_n8192_e3	node09	80.2527	0.3651	2.1210	2.8394	27.6807	52428800
T1/math	T1_selected_fusion_auf_n8192_e3	node09	80.1810	0.3648	2.1191	2.8402	26.8966	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_restore4	node08	71.1581	0.3238	1.8806	2.3407	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_delta32	node08	58.2957	0.2652	1.5407	2.1151	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_fusion_remove	node08	38.6362	0.1758	1.0211	1.4086	31.6433	52428800
T1/math	ar	node08	37.8373	0.1722	1.0000	1.0000	—	—
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_delta2	node09	29.2849	0.1332	0.7740	1.0780	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_delta1	node08	29.0863	0.1323	0.7687	1.0377	31.6433	52428800
T1/math	T1_sweep_auf_n8192_a32_e3 / selected8k_radial	node08	28.2457	0.1285	0.7465	1.0155	31.6433	52428800
T3/chat	ar	historical	64.7250	—	1.0000	1.0000	—	—
T3/chat	T3_fivew_decay7_n4096_a32	historical	54.1219	—	0.8362	1.2650	4.9464	62914560
T3/chat	T3_fivew_auf_n4096_a32	historical	54.0551	—	0.8352	1.2580	5.1403	62914560
T3/code	T3_fivew_decay7_n4096_a32	historical	63.5709	—	1.0077	1.4733	4.9464	62914560
T3/code	ar	historical	63.0876	—	1.0000	1.0000	—	—
T3/code	T3_fivew_auf_n4096_a32	historical	59.2704	—	0.9395	1.4663	5.1403	62914560
T3/code	T3_ba28_auf_n4096_a32	historical	48.0659	—	0.7619	1.1120	4.4572	1003520
T3/code	T3_fusion_auf_n4096_a32	historical	47.8449	—	0.7584	1.0959	4.6463	62914560
T3/code	T3_ba28_decay7_n4096_a32	historical	47.2628	—	0.7492	1.0989	4.4416	1003520
T3/code	T3_fusion_decay7_n4096_a32	historical	46.5790	—	0.7383	1.0818	4.5926	62914560
T3/gsm	T3_fivew_decay7_n4096_a32	historical	97.8498	—	1.5182	2.3818	4.9464	62914560
T3/gsm	T3_fivew_auf_n4096_a32	historical	96.3963	—	1.4957	2.3962	5.1403	62914560
T3/gsm	ar	historical	64.4510	—	1.0000	1.0000	—	—
T3/gsm	initial	historical	41.3441	—	0.6415	1.0087	—	—
T3/math	T3_fivew_auf_n4096_a32	historical	120.2144	—	1.9422	2.9048	5.1403	62914560
T3/math	T3_fivew_decay7_n4096_a32	historical	115.6398	—	1.8683	2.8995	4.9464	62914560
T3/math	ar	historical	61.8956	—	1.0000	1.0000	—	—
T3/math	T3_ba28_auf_n4096_a32	historical	59.1095	—	0.9550	1.3343	4.4572	1003520
T3/math	T3_fusion_auf_n4096_a32	historical	57.3693	—	0.9269	1.3182	4.6463	62914560
T3/math	T3_ba28_decay7_n4096_a32	historical	56.7761	—	0.9173	1.2679	4.4416	1003520
T3/math	T3_fusion_decay7_n4096_a32	historical	55.1158	—	0.8905	1.2317	4.5926	62914560
T3/math	initial	historical	42.3190	—	0.6837	1.0084	—	—
T4/chat	ar	historical	40.9593	—	1.0000	1.0000	—	—
T4/code	ar	historical	40.7880	—	1.0000	1.0000	—	—
T4/gsm	ar	historical	41.1094	—	1.0000	1.0000	—	—
T4/math	ar	historical	41.0048	—	1.0000	1.0000	—	—

G.1 COMPLETE FITS, INCLUDING THOSE WITHOUT A COMPLETE EVALUATION

A completed optimizer loop does not imply that a full inference cohort exists. The following training inventory reports those fits without inventing downstream performance. Setup, interface initialization and export durations remain separate from optimizer-loop time. Each checkpoint may serve multiple workload evaluations, so fitting costs are not multiplied by the number of task rows.

Table 31: Completed fits, including fits without completed evaluation

Fit	Examples	Epochs	LR	Updates	Fit min	GPU h	Setup / initialization / export seconds
T1_ba128_auf_n4096_a32	4096	1	0.0001	512	4.0568	0.1352	10.2284 / 1.7732 / 0.9332
T1_ba128_decay7_n4096_a32	4096	1	0.0001	512	4.0384	0.1346	8.5410 / 1.7643 / 1.1305

1674
1675
1676
1677
1678
1679
1680
1681
1682
1683
1684
1685
1686
1687
1688
1689
1690
1691
1692
1693
1694
1695
1696
1697
1698
1699
1700
1701
1702
1703
1704
1705
1706
1707
1708
1709
1710
1711
1712
1713
1714
1715
1716
1717
1718
1719
1720
1721
1722
1723
1724
1725
1726
1727

<i>Continued</i>								
Fit	Examples	Epochs	LR	Updates	Fit min	GPU h	Setup / initialization / export seconds	
T1_ba28_auf_n4096_a32	4096	1	0.0001	512	4.3310	0.1444	9.9652 / 1.7694 / 0.9785	
T1_ba28_decay7_n4096_a32	4096	1	0.0001	512	4.1859	0.1395	11.0379 / 1.7740 / 0.9926	
T1_ba56_auf_n4096_a32	4096	1	0.0001	512	4.1549	0.1385	10.7209 / 1.7698 / 0.9764	
T1_ba56_decay7_n4096_a32	4096	1	0.0001	512	4.0593	0.1353	11.2965 / 1.7753 / 1.0223	
T1_body128_auf_n4096_a32	4096	1	0.0001	512	5.4524	0.1817	11.9674 / 0.0000 / 2.2291	
T1_body128_decay7_n4096_a32	4096	1	0.0001	512	5.3792	0.1793	11.2803 / 0.0000 / 2.2115	
T1_fivew_auf_n4096_a32	4096	1	0.0001	512	4.7556	0.1585	5.8439 / 0.0176 / 1.4577	
T1_fivew_decay7_n4096_a32	4096	1	0.0001	512	4.5598	0.1520	9.5934 / 0.0190 / 1.2206	
T1_fusion_auf_n4096_a32	4096	1	0.0001	512	4.8197	0.1607	10.1458 / 0.0948 / 1.0784	
T1_fusion_decay7_n4096_a32	4096	1	0.0001	512	4.6981	0.1566	9.9911 / 0.0968 / 0.9068	
T1_joint_auf_n4096_a32	4096	1	0.0001	512	5.5329	0.1844	9.2628 / 0.0250 / 2.6514	
T1_joint_decay7_n4096_a32	4096	1	0.0001	512	5.5666	0.1856	7.6109 / 0.0229 / 2.4088	
T1_mse100_n4096_e3_lr0.001	4096	3	0.001	1536	18.3032	0.6101	4.4681 / 0.0264 / 1.2480	
T1_mse100_n8192_e3_lr0.001	8192	3	0.001	3072	53.9014	1.7967	3.4497 / 0.0223 / 1.3005	
T1_priority_auf_n16384_e3_lr1e4	16384	3	0.0001	6144	59.0507	1.9684	10.4315 / 0.0208 / 1.5610	
T1_priority_auf_n4096_plus2_lr1e4	4096	2	0.0001	1024	8.9126	0.2971	5.6374 / 0.0183 / 1.2830	
T1_priority_decay7_n4096_plus2_lr1e4	4096	2	0.0001	1024	8.7649	0.2922	9.0154 / 0.0202 / 1.3091	
T1_selected_ba128_auf_n8192_e3	8192	3	0.0001	3072	29.4859	0.9829	10.5382 / 1.7711 / 1.0282	
T1_selected_ba128_decay7_n8192_e3	8192	3	0.0001	3072	24.0569	0.8019	7.5216 / 1.8048 / 0.8727	
T1_selected_ba28_auf_n8192_e3	8192	3	0.0001	3072	29.7632	0.9921	7.2423 / 1.7154 / 1.0132	
T1_selected_ba28_decay7_n8192_e3	8192	3	0.0001	3072	28.1235	0.9374	7.0466 / 1.8259 / 0.8988	
T1_selected_ba56_auf_n8192_e3	8192	3	0.0001	3072	25.2175	0.8406	8.1886 / 1.7887 / 1.1010	
T1_selected_ba56_decay7_n8192_e3	8192	3	0.0001	3072	24.6055	0.8202	6.9974 / 1.7860 / 0.9206	
T1_selected_body128_auf_n8192_e3	8192	3	0.0001	3072	31.7912	1.0597	8.2654 / 0.0000 / 1.8327	

1728	<i>Continued</i>							
1729	Fit	Examples	Epochs	LR	Updates	Fit min	GPU h	Setup / initialization / export seconds
1730								
1731								
1732	T1_selected_body128_decay7_ n8192_e3	8192	3	0.0001	3072	31.7757	1.0592	10.8394 / 0.0000 / 1.8358
1733								
1734	T1_selected_fusion_auf_n8192_e3	8192	3	0.0001	3072	26.8966	0.8966	11.9389 / 0.0983 / 1.3188
1735								
1736	T1_selected_fusion_decay7_n8192_ e3	8192	3	0.0001	3072	27.6807	0.9227	8.5079 / 0.0939 / 1.1943
1737								
1738	T1_selected_joint_auf_n8192_e3	8192	3	0.0001	3072	36.3991	1.2133	9.5867 / 0.0255 / 2.0476
1739								
1740	T1_selected_joint_decay7_n8192_e3	8192	3	0.0001	3072	33.2872	1.1096	5.7355 / 0.0236 / 2.1655
1741								
1742	T1_sweep_auf_n4096_a32_e3	4096	3	0.0001	1536	13.2874	0.4429	10.4930 / 0.0302 / 1.4725
1743								
1744	T1_sweep_auf_n4096_a8_e3	4096	3	0.0001	1536	10.3192	0.3440	9.1982 / 0.0188 / 1.2071
1745								
1746	T1_sweep_auf_n8192_a32_e3	8192	3	0.0001	3072	31.6433	1.0548	8.5263 / 0.0194 / 1.3958
1747								
1748	T1_sweep_decay7_n4096_a32_e3	4096	3	0.0001	1536	13.0235	0.4341	8.7769 / 0.0246 / 1.3339
1749								
1750	T1_sweep_decay7_n8192_a32_e3	8192	3	0.0001	3072	28.2368	0.9412	16.5740 / 0.0197 / 1.5714
1751								
1752	T2_body128_auf_n4096_a32	4096	1	0.0001	512	3.7824	0.1261	9.0496 / 0.0000 / 1.2615
1753								
1754	T2_body128_decay7_n4096_a32	4096	1	0.0001	512	3.7345	0.1245	10.2005 / 0.0000 / 1.4344
1755								
1756	T2_fivew_auf_n4096_a32	4096	1	0.0001	512	5.5500	0.1850	7.0695 / 0.0179 / 1.5741
1757								
1758	T2_fivew_decay7_n4096_a32	4096	1	0.0001	512	5.3331	0.1778	11.5667 / 0.0226 / 1.5821
1759								
1760	T3_ba128_auf_n4096_a32	4096	1	0.0001	512	4.4946	0.1498	12.0737 / 3.8758 / 1.3747
1761								
1762	T3_ba128_decay7_n4096_a32	4096	1	0.0001	512	4.4919	0.1497	19.2670 / 3.8591 / 1.4384
1763								
1764	T3_ba28_auf_n4096_a32	4096	1	0.0001	512	4.4572	0.1486	13.2571 / 3.9048 / 1.3275
1765								
1766	T3_ba28_decay7_n4096_a32	4096	1	0.0001	512	4.4416	0.1481	10.0451 / 3.8592 / 1.4272
1767								
1768	T3_ba56_auf_n4096_a32	4096	1	0.0001	512	4.4592	0.1486	10.1981 / 3.8603 / 1.3467
1769								
1770	T3_ba56_decay7_n4096_a32	4096	1	0.0001	512	4.4726	0.1491	13.5962 / 3.8715 / 1.4131
1771								
1772	T3_fivew_auf_n4096_a32	4096	1	0.0001	512	5.1403	0.1713	9.0185 / 0.0197 / 2.3370
1773								
1774	T3_fivew_decay7_n4096_a32	4096	1	0.0001	512	4.9464	0.1649	10.2386 / 0.0197 / 1.6812
1775								
1776	T3_fusion_auf_n4096_a32	4096	1	0.0001	512	4.6463	0.1549	6.2923 / 0.0889 / 1.4870
1777								
1778	T3_fusion_decay7_n4096_a32	4096	1	0.0001	512	4.5926	0.1531	8.7555 / 0.0986 / 1.3900
1779								
1780								
1781								

Continued

Fit	Examples	Epochs	LR	Updates	Fit min	GPU h	Setup / initialization / export seconds
T3_joint_auf_n4096_a32	4096	1	0.0001	512	6.2167	0.2072	9.9182 / 0.0257 / 2.9022
T3_joint_decay7_n4096_a32	4096	1	0.0001	512	6.1806	0.2060	15.7252 / 0.0253 / 2.7877

G.2 SEPARATE LoRA COVERAGE CONTROLS

For completeness, the Phase 2 inventory also records six same-model LoRA-target coverage controls. They are not additional cross-model transfer pairs. All use 4,096 examples, one epoch, 512 updates, LR 10^{-4} and 32,768,000 trainable parameters. Prompt and response features receive 50% or 100% coverage. Evaluation has 128 prompts per domain, with response caps 2,048/8,192/2,048 for GSM8K/KiCad/NanoCoder. Fitting seconds exclude collection, setup and export. The mean-throughput change versus AUF also changes feature backend and supervision structure; it is not an isolated loss comparison.

Table 32: Separate LoRA coverage controls recorded alongside Phase 2; not cross-model transfer.

Domain	Coverage	Mean tok/s	$\times$ native	Gain / native	Change / AUF	Observed $\times$ AR	τ	Fit seconds	Fit GPU h
math	50%	128.54	1.0111	+1.11%	-26.51%	3.312	4.4665	24.00	0.0133
math	100%	124.80	0.9817	-1.83%	-28.65%	3.215	4.4696	34.12	0.0190
kicad	100%	196.83	1.2050	+20.50%	-36.42%	5.123	7.2064	381.61	0.2120
kicad	50%	194.45	1.1905	+19.05%	-37.19%	5.061	7.2083	181.75	0.1010
nanocoder	100%	148.15	0.9863	-1.37%	-8.73%	3.798	4.9742	37.45	0.0208
nanocoder	50%	146.78	0.9772	-2.28%	-9.57%	3.763	4.9573	29.13	0.0162

H INDEPENDENT FIVE-MAP CONFIRMATION (SUPPLEMENTARY ONLY)

The supplied RelaySpec report’s Appendix J ([RelaySpec](#)) independently trains five rectangular maps for T1 using 16,384 newly generated NuminaMath responses capped at 4,096, 25% segment-stratified positions, three epochs, Xavier seed 42 and normalized layer-plus-context MSE. It uses batches of 2,048 positions (7,911 updates), rather than our eight-example batches. It reports 5,398,695 selected positions per epoch and no validation-based checkpoint selection. Its NuminaMath evaluation cohort and vLLM runtime differ from our four-workload Transformers study; neither its weights nor its results are pooled into our canonical curves.

Assignment	AR pooled TPS	Native pooled TPS	Mapped pooled TPS	Mapped/native [95% CI]
Original	26.05	163.57	167.83	1.026 [1.018,1.034]
Reversed GPUs	26.05	163.67	167.97	1.026 [1.018,1.034]

This is single-request, not a concurrent-serving result. It uses BF16 batch-invariant vLLM 0.28 on L40S, greedy nonthinking decoding, no prefix cache, 128 prompts and cap 2,048. All arms return the same 124,899 tokens, with 104 natural stops and 24 cap stops; the AR measurement is reused across the two speculative passes. Native/mapped acceptance in the original comparison is 7.2741/7.0785, including the bonus. Paired request intervals are conditional on one trained checkpoint, not training-seed uncertainty. Source-reported fitting is 14.4 minutes on one GPU, with generation 96.4 minutes on two GPUs and capture/validation 52.0 minutes on two GPUs. These times have different boundaries from our fitting-only values. The independent report’s other transfer and short-context scaling campaigns are not included in this manuscript.

I COMPLETE OPTIMIZED CONCURRENT-SERVING MEASUREMENTS

All 52 non-AR method/workload/concurrency cells are complete: 32 for T1, eight for T2, four for T3 and eight for T4. Each contains 128 requests, for 6,656 requests total. T3 has no native arm; all 28 concurrent AR cells are complete, adding 3,584 requests for 10,240 requests overall. All retained cells have zero observed JIT events and use vLLM 0.28.0+cu129, PyTorch 2.13.0+cu129 and Transformers 5.16.1.

Use one vLLM engine per arm on an L40S, with the same selected checkpoint as standalone. BF16, FlashAttention, 15 draft proposals, mode-3 compilation, automatically supported async scheduling/graphs, 32 engine slots, 32,768 batched tokens and 80% GPU-memory fraction are fixed. T1–T3 use the V1 model runner; T4 uses V2 for its tokenizer bridge. The published DFlash integration command supplies the 32,768-token budget, FlashAttention and 15 proposals; the complete profile is an adapted version-pinned experiment, not an exact reproduction of every upstream setting. Prefix caching is enabled and reset before every measured cell. No numerical-output-equality certificate is assumed for the optimized profile.

Aggregate throughput is total returned tokens divided by measured workload wall time, including workload drain. Per-request latency starts after acquisition of the client concurrency semaphore; it includes the actual inference and local HTTP streaming, not waiting for a client slot. TTFT is time to first returned token; TPOT excludes that first-token latency and is defined only when more than one token is returned. vLLM acceptance is 1+ accepted draft tokens divided by verification steps using timed counter deltas. Allocation and reservation memory snapshots are not peak memory.

Warmup precedes timing. If observed JIT enters a measured cell, preserve the complete cold attempt, reset prefix cache and repeat in the same engine; accept the first zero-observed-JIT attempt, not the fastest one. Stop after three contaminated attempts. All attempt durations count toward experiment cost. This excludes detected compilation contamination, not every possible source of system noise.

Table 33: Optimized vLLM aggregate throughput and acceptance. No native arm exists for T3. All 28 optimized AR measurements are complete.

Pair	Task	Clients	AR TPS	Native TPS	Mapped TPS	Map/native	Map/AR	Native tau	Mapped tau
T1	math	1	46.048	276.089	292.228	1.058	6.346	8.102	7.917
T1	math	8	322.807	1752.388	1840.940	1.051	5.703	8.138	7.943
T1	math	16	569.359	2524.306	2716.261	1.076	4.771	8.110	7.930
T1	math	32	848.570	2866.964	3317.430	1.157	3.909	8.020	7.827
T1	gsm	1	45.963	216.540	227.948	1.053	4.959	6.454	6.248
T1	gsm	8	331.355	1301.710	1448.262	1.113	4.371	6.436	6.229
T1	gsm	16	604.435	2207.429	2262.026	1.025	3.742	6.450	6.254
T1	gsm	32	981.930	2251.468	2612.562	1.160	2.661	6.458	6.256
T1	code	1	45.813	192.432	176.516	0.917	3.853	5.719	4.826
T1	code	8	282.082	1106.866	1045.046	0.944	3.705	5.716	4.893
T1	code	16	505.138	1748.194	1548.474	0.886	3.065	5.906	4.893
T1	code	32	781.997	1940.519	1826.453	0.941	2.336	5.680	4.843
T1	chat	1	45.991	114.826	106.684	0.929	2.320	3.325	2.851
T1	chat	8	317.613	705.513	682.439	0.967	2.149	3.305	2.908
T1	chat	16	555.385	1091.138	1065.927	0.977	1.919	3.312	2.898
T1	chat	32	815.112	1349.441	1273.818	0.944	1.563	3.355	2.917

Table 34: Optimized vLLM aggregate throughput and acceptance. No native arm exists for T3. All 28 optimized AR measurements are complete.

Pair	Task	Clients	AR TPS	Native TPS	Mapped TPS	Map/native	Map/AR	Native tau	Mapped tau
T2	math	16	959.420	4482.419	3710.758	0.828	3.868	8.109	7.534
T2	gsm	16	1021.542	3160.992	3078.871	0.974	3.014	6.366	6.012
T2	code	16	825.292	2840.151	2002.791	0.705	2.427	6.163	4.537
T2	chat	16	941.994	1888.688	1431.215	0.758	1.519	3.393	2.788
T3	math	16	1126.927	—	2856.085	—	2.534	—	4.713
T3	gsm	16	727.302	—	2241.589	—	3.082	—	3.853
T3	code	16	1213.665	—	2077.617	—	1.712	—	3.529

Continued

Pair	Task	Clients	AR TPS	Native TPS	Mapped TPS	Map/native	Map/AR	Native tau	Mapped tau
T3	chat	16	1072.819	—	1899.952	—	1.771	—	2.989
T4	math	16	558.218	1894.999	1046.944	0.552	1.876	5.091	4.681
T4	gsm	16	394.436	1354.501	853.135	0.630	2.163	4.211	3.439
T4	code	16	585.584	1296.828	636.973	0.491	1.088	4.387	2.987
T4	chat	16	520.087	1440.038	524.399	0.364	1.008	3.986	2.043

Table 35: Completed concurrent AR comparisons. Rates use aggregate returned tokens over workload wall time, including drain.

Pair	Task	Clients	AR TPS	Native / AR	Mapped / AR
T1	math	1	46.048	5.996	6.346
T1	math	8	322.807	5.429	5.703
T1	math	16	569.359	4.434	4.771
T1	math	32	848.570	3.379	3.909
T1	gsm	1	45.963	4.711	4.959
T1	gsm	8	331.355	3.928	4.371
T1	gsm	16	604.435	3.652	3.742
T1	gsm	32	981.930	2.293	2.661
T1	code	1	45.813	4.200	3.853
T1	code	8	282.082	3.924	3.705
T1	code	16	505.138	3.461	3.065
T1	code	32	781.997	2.481	2.336
T1	chat	1	45.991	2.497	2.320
T1	chat	8	317.613	2.221	2.149
T1	chat	16	555.385	1.965	1.919
T1	chat	32	815.112	1.656	1.563
T2	math	16	959.420	4.672	3.868
T2	gsm	16	1021.542	3.094	3.014
T2	code	16	825.292	3.441	2.427
T2	chat	16	941.994	2.005	1.519
T3	math	16	1126.927	—	2.534
T3	gsm	16	727.302	—	3.082
T3	code	16	1213.665	—	1.712
T3	chat	16	1072.819	—	1.771
T4	math	16	558.218	3.395	1.876
T4	gsm	16	394.436	3.434	2.163
T4	code	16	585.584	2.215	1.088
T4	chat	16	520.087	2.769	1.008

Table 36: Per-request serving latency in seconds; queueing before client-slot acquisition is excluded. Percentiles use linear interpolation.

Pair	Task	Clients	Arm	Mean latency	P50	P95	Mean TTFT	Mean TPOT
T1	math	1	native	2.502	1.978	6.067	0.061	0.004
T1	math	1	mapped	2.395	1.957	6.064	0.057	0.003
T1	math	8	native	3.047	2.470	7.555	0.108	0.004
T1	math	8	mapped	3.017	2.384	7.191	0.101	0.004
T1	math	16	native	3.749	3.031	9.586	0.140	0.005
T1	math	16	mapped	3.644	2.938	8.815	0.133	0.005
T1	math	32	native	6.078	4.710	14.788	0.234	0.009
T1	math	32	mapped	5.789	4.567	13.977	0.222	0.008
T1	gsm	1	native	1.270	1.221	2.139	0.061	0.004
T1	gsm	1	mapped	1.208	1.155	1.888	0.057	0.004
T1	gsm	8	native	1.582	1.465	2.461	0.102	0.005
T1	gsm	8	mapped	1.463	1.408	2.312	0.095	0.005
T1	gsm	16	native	1.876	1.807	3.016	0.132	0.006
T1	gsm	16	mapped	1.803	1.723	2.895	0.128	0.006
T1	gsm	32	native	2.915	2.859	4.765	0.230	0.010
T1	gsm	32	mapped	2.800	2.794	4.533	0.212	0.010
T1	code	1	native	2.903	2.752	6.363	0.080	0.005
T1	code	1	mapped	3.193	2.989	6.603	0.076	0.006
T1	code	8	native	3.861	3.601	9.003	0.147	0.006

1944

Continued

1945

1946

1947

1948

1949

1950

1951

1952

1953

1954

1955

1956

1957

1958

1959

1960

1961

1962

1963

1964

1965

1966

1967

1968

1969

1970

1971

1972

1973

1974

1975

1976

1977

1978

1979

1980

1981

1982

1983

1984

1985

1986

1987

1988

1989

1990

1991

1992

1993

1994

1995

1996

1997

Pair	Task	Clients	Arm	Mean latency	P50	P95	Mean TTFT	Mean TPOT
T1	code	8	mapped	4.209	3.835	9.494	0.143	0.007
T1	code	16	native	4.802	4.622	10.495	0.265	0.008
T1	code	16	mapped	5.338	4.922	10.796	0.219	0.009
T1	code	32	native	7.802	7.349	18.337	0.531	0.013
T1	code	32	mapped	8.982	8.130	22.209	0.502	0.015
T1	chat	1	native	6.323	5.743	15.228	0.066	0.009
T1	chat	1	mapped	6.640	5.882	15.500	0.062	0.010
T1	chat	8	native	7.741	6.645	18.624	0.112	0.012
T1	chat	8	mapped	8.119	7.691	18.666	0.104	0.012
T1	chat	16	native	9.075	8.443	21.622	0.138	0.014
T1	chat	16	mapped	9.867	8.848	25.096	0.133	0.015
T1	chat	32	native	14.190	13.086	33.547	0.276	0.022
T1	chat	32	mapped	15.388	13.892	35.306	0.244	0.024
T2	math	16	native	2.348	1.923	5.322	0.084	0.003
T2	math	16	mapped	2.838	2.239	6.847	0.093	0.004
T2	gsm	16	native	1.121	1.024	1.811	0.076	0.004
T2	gsm	16	mapped	1.299	1.199	2.108	0.085	0.005
T2	code	16	native	2.676	2.346	7.038	0.139	0.005
T2	code	16	mapped	3.545	3.163	8.419	0.149	0.007
T2	chat	16	native	5.590	4.986	13.044	0.085	0.009
T2	chat	16	mapped	7.293	6.601	16.487	0.093	0.011
T3	math	16	mapped	2.655	1.669	7.617	0.076	0.006
T3	gsm	16	mapped	1.115	0.994	1.893	0.072	0.006
T3	code	16	mapped	2.968	2.820	5.239	0.120	0.008
T3	chat	16	mapped	3.914	3.593	7.953	0.074	0.008
T4	math	16	native	5.302	3.337	13.392	0.136	0.009
T4	math	16	mapped	8.177	5.049	23.501	0.191	0.015
T4	gsm	16	native	1.864	1.723	3.181	0.131	0.009
T4	gsm	16	mapped	3.386	3.043	5.481	0.167	0.016
T4	code	16	native	4.210	3.762	8.171	0.237	0.011
T4	code	16	mapped	8.680	7.923	16.750	0.299	0.024
T4	chat	16	native	5.235	4.885	9.648	0.139	0.011
T4	chat	16	mapped	13.882	13.129	27.477	0.188	0.029
T1	math	1	ar	14.838	11.362	43.707	0.065	0.022
T1	math	8	ar	17.042	13.094	49.249	0.080	0.024
T1	math	16	ar	17.968	13.380	51.480	0.092	0.025
T1	math	32	ar	19.009	14.184	54.407	0.139	0.027
T1	gsm	1	ar	6.001	5.760	10.961	0.070	0.022
T1	gsm	8	ar	6.488	6.249	10.623	0.078	0.023
T1	gsm	16	ar	6.582	6.376	10.549	0.086	0.024
T1	gsm	32	ar	7.016	6.893	10.823	0.132	0.026
T1	code	1	ar	12.253	11.558	25.759	0.079	0.022
T1	code	8	ar	13.517	13.541	26.959	0.106	0.025
T1	code	16	ar	16.162	13.421	54.928	0.170	0.027
T1	code	32	ar	17.662	16.420	55.572	0.378	0.030
T1	chat	1	ar	15.795	13.548	38.868	0.064	0.022
T1	chat	8	ar	17.324	15.425	41.174	0.082	0.024
T1	chat	16	ar	18.400	16.324	44.619	0.092	0.025
T1	chat	32	ar	20.377	17.376	47.914	0.178	0.028
T2	math	16	ar	10.919	8.051	30.842	0.071	0.015
T2	gsm	16	ar	3.945	3.762	6.402	0.068	0.014
T2	code	16	ar	9.176	6.894	33.883	0.111	0.016
T2	chat	16	ar	10.772	9.530	28.984	0.063	0.015
T3	math	16	ar	5.777	2.867	24.598	0.055	0.012
T3	gsm	16	ar	2.356	1.882	3.534	0.057	0.012
T3	code	16	ar	4.467	4.311	6.426	0.085	0.012
T3	chat	16	ar	5.838	5.293	13.996	0.051	0.012
T4	math	16	ar	15.113	7.693	50.656	0.090	0.024
T4	gsm	16	ar	5.745	4.272	8.588	0.088	0.023
T4	code	16	ar	9.666	8.474	17.620	0.165	0.025
T4	chat	16	ar	11.914	10.504	23.077	0.091	0.024

1998

1999

2000

2001

2002

2003

2004

2005

2006

2007

2008

2009

2010

2011

2012

2013

2014

2015

2016

2017

2018

2019

2020

2021

2022

2023

2024

2025

2026

2027

2028

2029

2030

2031

2032

2033

2034

2035

2036

2037

2038

2039

2040

2041

2042

2043

2044

2045

2046

2047

2048

2049

2050

2051

Table 37: Serving output, acceptance counters and post-cell worker memory snapshots. Memory is not a peak measurement. Acceptance equals 1 + accepted / steps.

Pair	Task	Clients	Arm	Tokens	Wall s	Accepted	Steps	Allocated GiB	Reserved GiB
T1	math	1	native	88428	320.288	77626	10930	30.591	30.828
T1	math	1	mapped	89603	306.620	78326	11324	30.919	31.229
T1	math	8	native	87904	50.162	77207	10816	30.591	31.039
T1	math	8	mapped	91446	49.674	79978	11520	30.919	31.439
T1	math	16	native	88970	35.245	78111	10986	30.591	31.387
T1	math	16	mapped	90527	33.328	79151	11421	30.919	31.787
T1	math	32	native	91908	32.058	80585	11480	30.591	32.293
T1	math	32	mapped	91230	27.500	79607	11660	30.919	32.709
T1	gsm	1	native	35201	162.561	29841	5471	30.591	32.293
T1	gsm	1	mapped	35259	154.680	29612	5642	30.919	32.709
T1	gsm	8	native	35953	27.620	30460	5603	30.591	32.293
T1	gsm	8	mapped	34857	24.068	29257	5595	30.919	32.709
T1	gsm	16	native	35186	15.940	29823	5472	30.591	32.293
T1	gsm	16	mapped	35313	15.611	29663	5646	30.919	32.709
T1	gsm	32	native	34926	15.513	29613	5426	30.591	32.293
T1	gsm	32	mapped	35005	13.399	29405	5595	30.919	32.709
T1	code	1	native	71508	371.602	59126	12530	30.591	32.293
T1	code	1	mapped	72148	408.733	57212	14953	30.919	32.709
T1	code	8	native	73176	66.111	60502	12830	30.591	32.293
T1	code	8	mapped	74255	71.054	59112	15184	30.919	32.709
T1	code	16	native	72589	41.522	60428	12318	30.591	32.584
T1	code	16	mapped	73832	47.680	58766	15094	30.919	33.043
T1	code	32	native	70883	36.528	58501	12500	30.591	34.205
T1	code	32	mapped	76011	41.617	60352	15703	30.919	34.664
T1	chat	1	native	92932	809.332	65049	27978	30.591	34.205
T1	chat	1	mapped	90678	849.969	58892	31815	30.919	34.664
T1	chat	8	native	92311	130.842	64468	27963	30.591	34.205
T1	chat	8	mapped	91975	134.774	60366	31636	30.919	31.738
T1	chat	16	native	90406	82.855	63183	27333	30.591	31.494
T1	chat	16	mapped	92152	86.452	60372	31815	30.919	32.012
T1	chat	32	native	91738	67.982	64470	27378	30.591	32.451
T1	chat	32	mapped	92651	72.735	60911	31772	30.919	32.918
T2	math	16	native	90452	20.179	79433	11174	31.281	31.818
T2	math	16	mapped	91256	24.592	79175	12118	31.836	32.453
T2	gsm	16	native	35396	11.198	29928	5577	31.281	31.818
T2	gsm	16	mapped	34718	11.276	28942	5774	31.836	32.453
T2	code	16	native	66900	23.555	56195	10884	31.281	32.619
T2	code	16	mapped	61617	30.766	48058	13587	31.836	33.254
T2	chat	16	native	91027	48.196	64301	26865	31.281	32.619
T2	chat	16	mapped	88848	62.079	56987	31875	31.836	33.254
T3	math	16	mapped	68413	23.953	53847	14503	32.326	33.000
T3	gsm	16	mapped	23541	10.502	17337	6076	32.326	33.000
T3	code	16	mapped	51379	24.730	36737	14529	32.326	33.832
T3	chat	16	mapped	64619	34.011	42924	21581	32.326	33.832
T4	math	16	native	86317	45.550	69361	16953	30.900	31.859
T4	math	16	mapped	83062	79.338	65312	17743	30.891	31.996
T4	gsm	16	native	24508	18.094	18591	5789	30.900	31.859
T4	gsm	16	mapped	29169	34.190	20612	8450	30.891	31.996
T4	code	16	native	50067	38.607	38571	11388	30.900	32.824
T4	code	16	mapped	48159	75.606	31958	16085	30.891	32.961
T4	chat	16	native	65217	45.288	48774	16334	30.900	32.824
T4	chat	16	mapped	62812	119.779	32000	30692	30.891	32.961
T1	math	1	ar	87457	1899.243	—	—	30.953	31.951
T1	math	8	ar	90698	280.966	—	—	30.953	31.951
T1	math	16	ar	91439	160.600	—	—	30.953	31.951
T1	math	32	ar	88009	103.714	—	—	30.953	31.953
T1	gsm	1	ar	35305	768.117	—	—	30.953	31.953
T1	gsm	8	ar	35189	106.197	—	—	30.953	31.953
T1	gsm	16	ar	34876	57.700	—	—	30.953	31.953
T1	gsm	32	ar	34672	35.310	—	—	30.953	31.953
T1	code	1	ar	71854	1568.422	—	—	30.953	31.953
T1	code	8	ar	69802	247.453	—	—	32.289	32.779
T1	code	16	ar	76856	152.148	—	—	32.289	33.449
T1	code	32	ar	72849	93.158	—	—	32.289	34.367
T1	chat	1	ar	92981	2021.716	—	—	32.289	34.367
T1	chat	8	ar	91789	288.997	—	—	32.289	34.367
T1	chat	16	ar	92695	166.902	—	—	32.289	34.367

Continued

Pair	Task	Clients	Arm	Tokens	Wall s	Accepted	Steps	Allocated GiB	Reserved GiB
T1	chat	32	ar	91728	112.534	—	—	32.289	34.367
T2	math	16	ar	92944	96.875	—	—	32.146	33.045
T2	gsm	16	ar	35159	34.418	—	—	32.146	33.045
T2	code	16	ar	71057	86.099	—	—	32.146	33.178
T2	chat	16	ar	92068	97.737	—	—	32.146	33.178
T3	math	16	ar	61977	54.996	—	—	33.183	33.383
T3	gsm	16	ar	26014	35.768	—	—	33.183	33.385
T3	code	16	ar	45129	37.184	—	—	33.183	33.961
T3	chat	16	ar	64174	59.818	—	—	33.183	33.961
T4	math	16	ar	79647	142.681	—	—	32.048	32.328
T4	gsm	16	ar	31478	79.805	—	—	32.048	32.328
T4	code	16	ar	48480	82.789	—	—	32.048	33.260
T4	chat	16	ar	63593	122.274	—	—	32.048	33.260

Table 38: Draft acceptance survival across verification steps; L excludes the verifier bonus. All 15 positions are preserved in the numerical source.

Pair	Task	Clients	Arm	P(L>=1)	P(L>=4)	P(L>=8)	P(L>=12)	P(L>=15)
T1	math	1	native	0.913	0.631	0.421	0.293	0.216
T1	math	1	mapped	0.903	0.617	0.408	0.287	0.207
T1	math	8	native	0.912	0.640	0.423	0.297	0.218
T1	math	8	mapped	0.903	0.620	0.410	0.288	0.209
T1	math	16	native	0.913	0.635	0.423	0.295	0.213
T1	math	16	mapped	0.900	0.616	0.408	0.287	0.212
T1	math	32	native	0.909	0.624	0.415	0.293	0.214
T1	math	32	mapped	0.898	0.608	0.402	0.277	0.202
T1	gsm	1	native	0.883	0.532	0.292	0.165	0.097
T1	gsm	1	mapped	0.872	0.509	0.281	0.155	0.091
T1	gsm	8	native	0.886	0.528	0.293	0.163	0.097
T1	gsm	8	mapped	0.874	0.513	0.279	0.150	0.087
T1	gsm	16	native	0.884	0.528	0.296	0.162	0.096
T1	gsm	16	mapped	0.876	0.511	0.280	0.155	0.093
T1	gsm	32	native	0.884	0.537	0.295	0.163	0.097
T1	gsm	32	mapped	0.875	0.515	0.278	0.153	0.090
T1	code	1	native	0.868	0.459	0.226	0.128	0.087
T1	code	1	mapped	0.828	0.391	0.155	0.075	0.046
T1	code	8	native	0.869	0.454	0.227	0.128	0.087
T1	code	8	mapped	0.825	0.400	0.161	0.078	0.051
T1	code	16	native	0.869	0.471	0.242	0.141	0.101
T1	code	16	mapped	0.830	0.392	0.162	0.082	0.054
T1	code	32	native	0.868	0.455	0.225	0.123	0.078
T1	code	32	mapped	0.828	0.390	0.159	0.078	0.048
T1	chat	1	native	0.712	0.214	0.067	0.027	0.015
T1	chat	1	mapped	0.640	0.158	0.042	0.018	0.011
T1	chat	8	native	0.706	0.210	0.065	0.028	0.018
T1	chat	8	mapped	0.641	0.164	0.048	0.021	0.012
T1	chat	16	native	0.710	0.210	0.064	0.028	0.017
T1	chat	16	mapped	0.639	0.162	0.047	0.022	0.014
T1	chat	32	native	0.711	0.214	0.068	0.030	0.018
T1	chat	32	mapped	0.638	0.163	0.049	0.023	0.015
T2	math	16	native	0.913	0.633	0.428	0.294	0.211
T2	math	16	mapped	0.900	0.589	0.382	0.254	0.178
T2	gsm	16	native	0.881	0.533	0.289	0.152	0.091
T2	gsm	16	mapped	0.871	0.501	0.259	0.134	0.079
T2	code	16	native	0.872	0.491	0.260	0.162	0.116
T2	code	16	mapped	0.808	0.356	0.138	0.066	0.037
T2	chat	16	native	0.706	0.214	0.072	0.037	0.025
T2	chat	16	mapped	0.629	0.151	0.041	0.015	0.009
T3	math	16	mapped	0.810	0.412	0.185	0.018	0.008
T3	gsm	16	mapped	0.831	0.328	0.059	0.002	0.000
T3	code	16	mapped	0.755	0.272	0.071	0.001	0.000
T3	chat	16	mapped	0.670	0.195	0.050	0.001	0.000
T4	math	16	native	0.853	0.456	0.221	0.019	0.000
T4	math	16	mapped	0.690	0.357	0.190	0.100	0.042
T4	gsm	16	native	0.868	0.383	0.079	0.003	0.000
T4	gsm	16	mapped	0.657	0.243	0.075	0.034	0.011

Continued

Pair	Task	Clients	Arm	P(L>=1)	P(L>=4)	P(L>=8)	P(L>=12)	P(L>=15)
T4	code	16	native	0.836	0.393	0.123	0.005	0.000
T4	code	16	mapped	0.673	0.176	0.042	0.014	0.004
T4	chat	16	native	0.804	0.331	0.100	0.005	0.000
T4	chat	16	mapped	0.488	0.069	0.013	0.003	0.001

I.1 EARLIER SERVING PROFILE, REPORTED SEPARATELY

These completed measurements use the earlier compilation-mode-0 profile: async scheduling disabled, FULL_DECODE_ONLY CUDA graphs, 8,192 batched tokens, capacity32, memory fraction0.8 and prefix caching enabled/reset per cell. They are retained for completeness, not combined with the optimized campaign or used for its headline ratios. The optimized tables above are the canonical serving results. The T3 table also repeats standalone rates to identify the shared checkpoint; its vLLM columns are the older serving measurements. Each cohort contains 128 prompts. These are distinct timing profiles, not independent repeats of the same configuration.

Table 39: Earlier serving profile: T1 vLLM concurrent serving — selected 16k / 25% mapper. Not pooled with optimized measurements.

Dataset	Clients	Native tok/s	Mapped tok/s	Mapped / native	Native τ	Mapped τ
math	1	184.17	189.84	1.0308 $\times$	8.0856	7.9319
math	8	1489.24	1552.06	1.0422 $\times$	8.0856	7.9319
math	16	2221.08	2325.92	1.0472 $\times$	8.0856	7.9319
math	32	2533.49	2716.26	1.0721 $\times$	8.0856	7.9319
gsm	1	144.40	148.03	1.0251 $\times$	6.4725	6.2847
gsm	8	1143.84	1170.43	1.0232 $\times$	6.4725	6.2847
gsm	16	1839.15	1904.40	1.0355 $\times$	6.4725	6.2847
gsm	32	2220.14	2337.06	1.0527 $\times$	6.4725	6.2847
code	1	131.26	117.00	0.8913 $\times$	5.8005	4.8945
code	8	1037.92	924.55	0.8908 $\times$	5.8005	4.8945
code	16	1559.34	1394.86	0.8945 $\times$	5.8005	4.8945
code	32	1851.73	1653.06	0.8927 $\times$	5.8005	4.8945
chat	1	76.17	69.90	0.9177 $\times$	3.3027	2.8734
chat	8	651.22	601.48	0.9236 $\times$	3.3027	2.8734
chat	16	1023.80	961.20	0.9389 $\times$	3.3027	2.8734
chat	32	1149.97	1103.12	0.9593 $\times$	3.3027	2.8734

Table 40: Earlier serving profile: T2 vLLM serving — selected 16k / 25% five-map MSE. Not pooled with optimized measurements.

Dataset	Native TPS	Mapped TPS	Mapped/native	Native τ	Mapped τ
math	3412.62	3049.92	0.8937 $\times$	8.0578	7.6492
gsm	2619.51	2289.25	0.8739 $\times$	6.3195	5.9828
code	2286.85	1741.66	0.7616 $\times$	6.0305	4.6646
chat	1387.87	1095.09	0.7890 $\times$	3.3130	2.7858

Table 41: Earlier serving profile: T3 descriptive non-AR throughput and acceptance. Not pooled with optimized measurements.

Dataset	TF mean TPS	TF pooled TPS	TF avg. τ	vLLM aggregate TPS, 16 clients	vLLM avg. τ	vLLM mean request latency (s)
math	145.43	176.56	4.7798	2279.89	4.5536	2.950
gsm	140.60	149.88	4.1593	1674.42	4.1339	1.375
code	118.57	125.29	3.4400	1698.40	3.3662	3.093
chat	104.43	108.34	2.9735	1670.18	3.0344	4.508

Table 42: Earlier serving profile: T4 completed non-AR evaluation. Not pooled with optimized measurements.

Engine	Dataset	Native TPS	Mapped TPS	Mapped/native	Native τ	Mapped τ
standalone	math	130.07	121.94	0.9374 $\times$	5.1211	4.6341
standalone	gsm	127.35	90.84	0.7133 $\times$	4.4245	3.2257
standalone	code	122.44	80.60	0.6583 $\times$	4.5230	3.0537
standalone	chat	114.42	60.52	0.5289 $\times$	3.9538	2.0640
serving	math	1660.77	992.81	0.5978 $\times$	5.3363	4.9853
serving	gsm	1247.78	815.41	0.6535 $\times$	4.5528	3.3694
serving	code	1240.05	607.94	0.4903 $\times$	4.3360	3.0181
serving	chat	1256.25	473.94	0.3773 $\times$	3.9403	2.0407

J REPRODUCIBILITY, ETHICS AND EVIDENCE BOUNDARIES

The manuscript’s source snapshot, table extraction, raw quarter-coverage result copies, figure generation and source-file hashes are packaged together. Numerical tables are generated from explicit completed-result tables and raw rows for every coverage setting in the scaling study. A build check verifies required evidence coverage, complete serving cells and citation/reference resolution. The draft does not claim that its data artifacts contain redistributable model weights or that all public-data generation metadata has been recovered.

No additional model training is performed to write this paper. These experiments optimize inference rather than measure improved task quality; they do not establish safety properties of generated content. Model and dataset releases retain their respective licenses. The paper has no author names or identifying acknowledgments. The remaining provenance/uncertainty limitations mean that a final submission needs a numerical and release audit; formatting a manuscript is not evidence that those scientific obligations have been fulfilled.